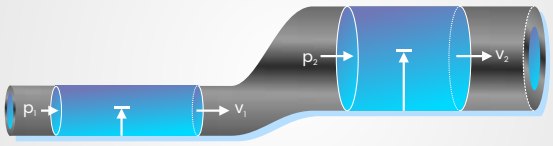
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Fluid Mechanics

(English Medium)

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

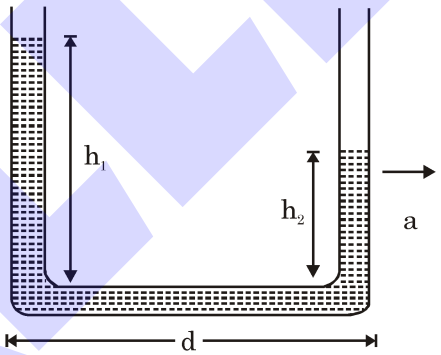
Fluid Statics :- Pressure and Pascal's law

1. Equal masses of water and a liquid of density 2 are mixed together, then the mixture has a density of
(A) 2/3 (B) 4/3 (C) 3/2 (D) 3 **FM0206**

2. Three liquids of relative density $d, 2d$ and $3d$ are mixed in equal proportions of weights. The relative density of the mixture is
(A) $\frac{11d}{7}$ (B) $\frac{18d}{11}$ (C) $\frac{13d}{9}$ (D) $\frac{23d}{18}$ **FM0207**

3. A bowl of soap water is at rest on a table in the dining compartment of a train, if the acceleration of the train is g/4 in forward direction, the angle made by its surface with horizontal is
(A) $\text{Tan}^{-1}\left(\frac{1}{2}\right)$ (B) $\text{Tan}^{-1}\left(\frac{1}{4}\right)$ (C) $\text{Tan}^{-1}\left(\frac{1}{5}\right)$ (D) $\text{Tan}^{-1}\left(\frac{1}{3}\right)$
FM0208

4. Fig. shows a U-tube of uniform cross-sectional area A accelerated with acceleration 'a' as shown. If d is the separation between the limbs, then the difference in the levels of the liquid in the U-tube is



(A) $\frac{ad}{g}$ (B) $\frac{g}{ad}$ (C) adg (D) ad + g
FM0209

5. **Statement-1 :** The free surface of a liquid at rest with respect to stationary container is always normal to the $\vec{g}$.
Statement-2 : Liquids at rest cannot have shear stress.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.
FM0096

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

6. An open rectangular tank 3 m wide, 2 m deep and 2m long is half filled with water. It is accelerated horizontally at 3.27 m/s^2 in the direction of its width. Select correct option regarding the depth of water at each end of tank. [g = 9.81 m/s^2]

(A) Depth of shallow end is 0.5 m

(B) Depth of shallow end is zero

(C) Depth of deeper end is 1 m

(D) Depth of deeper end is 2 m **FM0210**

7. A liquid kept in a cylindrical vessel of radius 0.3m is rotated with a speed 2r.ps. The difference in the height of the liquid at the centre of the vessel and at it's sides is

(A) 0.01m (B) 0.02m (C) 0.04m (D) 0.72m

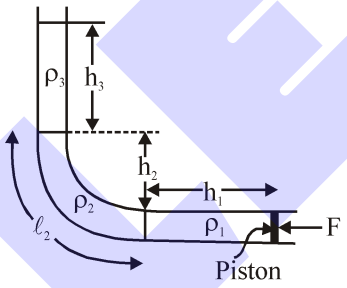
FM0211

8. The pressure at the bottom of a lake, due to water is 4.9×10^6 N/m^2. What is the depth of the lake?

(A) 500 m (B) 400 m (C) 300 m (D) 200m

FM0212

9. A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in equilibrium by the piston of area A, the value of $\frac{F}{A}$ is :

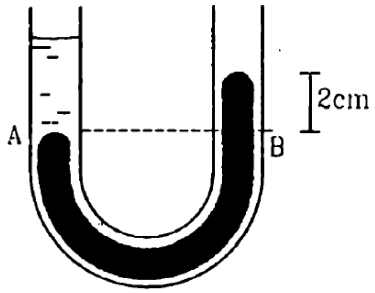


(A) $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$ (B) $(\rho_1 h_1 + \rho_2 \ell_2 + \rho_3 h_3)g$

(C) $(\rho_2 h_2 + \rho_3 h_3)g$ (D) $(\rho_2 \ell_2 + \rho_3 h_3)g$

FM0053

10. The liquids shown in figure in the two arms are mercury (specific gravity = 13.6) and water. If the difference of heights of the mercury columns is 2 cm, find the approximate height h of the water column.



(A) 22 cm (B) 25 cm (C) 27 cm (D) 29 cm

FM0213

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

11. Equal mass of three liquides are kept in three idential cylindrical vessel A, B, and C. The densities are ρ_A, ρ_B, ρ_C with $\rho_A < \rho_B < \rho_C$. The force on the base will be

(A) maximum in vessel A (B) maximum in vessel B

(C) maximum in vssel C (D) equal in all the vessels. **FM0214**

12. Suppose the pressure at the surface of mercury in a barometer tube is P_1 and the pressure at the surface of mercury in the cup is P_2.

(A) $P_1 = 0$, P_2 = atomspheric pressure

(B) P_1 = atomspheric pressure, $P_2 = 0$

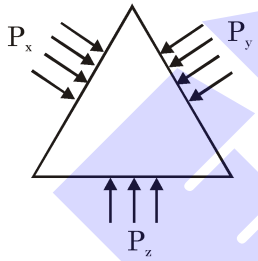
(C) $P_1 = P_2$ = atomspheric pressure

(D) $P_1 = P_2 = 0$ **FM0215**

13. A barometer kept in an elevator acceleration upward reads 76 cm. The air pressure in the elevator is

(A) 76 cm (B) < 76 cm (C) > 76 cm (D) zero **FM0216**

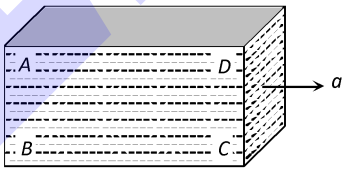
14. A horizontal triangular element of the liquid is shown in the fig., P_x, P_y and P_z represent the pressures on the element of the liquid. Then:



(A) $P_x = P_y \neq P_z$ (B) $P_x = P_y = P_z$

(C) $P_x \neq P_y \neq P_z$ (D) $P_x^2 + P_y^2 + P_z^2 =$ constant **FM0217**

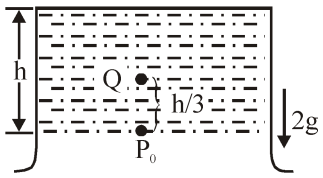
15. A closed rectangular tank is completely filled with water and is accelerated horizontally with an acceleration a towards right. Pressure is (i) maximum at, and (ii) minimum at



(A) (i) B (ii) D (B) (i) C (ii) D

(C) (i) B (ii) C (D) (i) B (ii) A **FM0218**

16. An inverted test tube having a liquid of density ρ accelerates down with a = 2g then, $P_Q - P_0$ is:

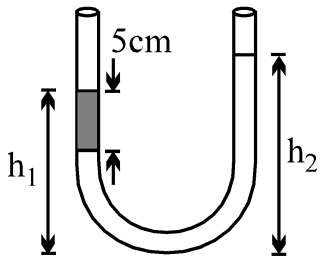


(A) $-\frac{\rho gh}{3}$ (B) $\frac{2\rho gh}{3}$ (C) $\frac{\rho gh}{3}$ (D) indeterminate

FM0118

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

17. An open-ended U-tube of uniform cross-sectional area contains water (density 1.0 gram/centimeter^3) standing initially 20 centimeters from the bottom in each arm. An immiscible liquid of density 4.0 grams/ centimeter^3 is added to one arm until a layer 5 centimeters high forms, as shown in the figure above. What is the ratio h_2/h_1 of the heights of the liquid in the two arms?



(A) 3/1 (B) 5/2 (C) 2/1 (D) 3/2

FM0054

18. A barometer reads 75.0 cm on a steel scale. The room temperature is 30 °C . The Scale is correctly graduated for 0°C . The coefficient of linear expansion of steel is $\alpha = 1.2 \times 10^{-5}$/°C and the coefficient of volume expansion of mercury is $\gamma = 1.8 \times 10^{-4}$/°C. Correct atmospheric pressure (in cm of Hg) is closest to :-

(A) 75 (B) 75.4 (C) 74 (D) 74.6 **FM0219**

19. A thin walled cylindrical metal vessel of linear coefficient of expansion 10^{-3} °C^{-1} contains benzene of volume expansion coefficient 10^{-3}°C^{-1}. If the vessel and its contents are now heated by 10°C, the pressure due to the liquid at the bottom.

(A) increases by 2% (B) decreases by 1%

(C) decreases by 2% (D) remains unchanged

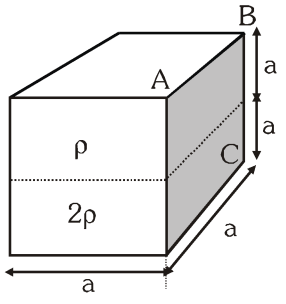
FM0058

20. The side of glass aquarium is 1 m high and 2 m long. When the aquarium is filled to the top with water, what is the total force against the side–

(A) 980×10^3 N (B) 9.8×10^3 N

(C) 0.98×10^3 N (D) 0.098×10^3 N **FM0220**

21. A cuboid ($a \times a \times 2a$) is filled with two immiscible liquids of density 2ρ & ρ as shown in the figure. Neglecting atmospheric pressure, ratio of force on base & side wall of the cuboid is :-

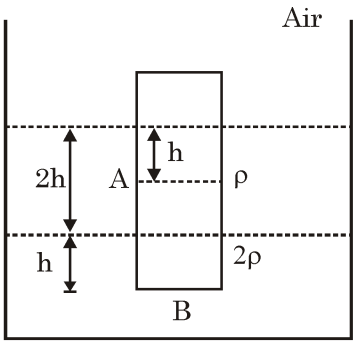


(A) 2 : 3 (B) 1 : 3 (C) 5 : 6 (D) 6 : 5

FM0056

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

22. A cylinder stands vertical in two immiscible liquids of densities r and 2r as shown. Find the difference in pressure at point A and B:



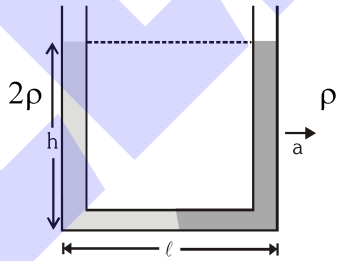
(A) 2 ρgh (B) 3 ρgh (C) 4 ρgh (D) None **FM0221**

23. Some liquid is filled in a cylindrical vessel of radius R. Let F_1 be the force applied by the liquid on the bottom of the cylinder. Now the same liquid is poured into a vessel of uniform square crss-section of side R. Let F_2 be the force applied by the liquid on the bottom of this new vessel.(Neglect atmosphere pressure) Then:

(A) $F_1 = \pi F_2$ (B) $F_1 = \dfrac{F_2}{\pi}$ (C) $F_1 = \sqrt{\pi} F_2$ (D) $F_1 = F_2$

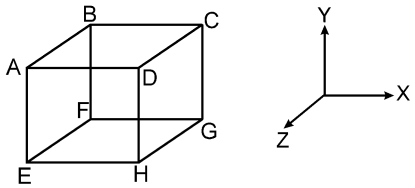
FM0119

24. A U–tube of base length 'ℓ' filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration 'a' on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height h is given by–



(A) $\dfrac{a}{2g}\ell$ (B) $\dfrac{3a}{2g}\ell$ (C) $\dfrac{a}{g}\ell$ (D) $\dfrac{2a}{3g}\ell$ **FM0222**

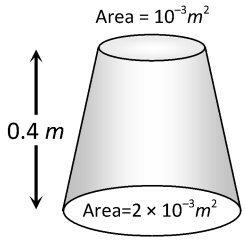
25. The cubical container ABCDEFGH which is completely filled with an ideal (nonviscous and incompressible) fluid, moves in a gravity free space with a acceleration of $\vec{a} = a_0(\hat{i} - \hat{j} + \hat{k})$ where a_0 is a positive constant. Then the only point in the container where pressure can be minimum is :



(A) H (B) C (C) E (D) F **FM0223**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

26. A uniformly tapering vessel kept in vacuum is filled with a liquid of density 900 kg/m^3. The force that acts on the base of the vessel due to the liquid is($g = 10\ ms^{-2}$)

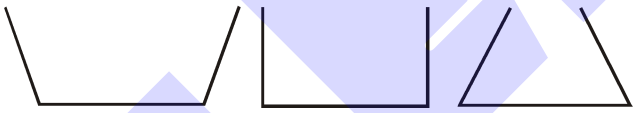


(A) 3.6 N (B) 7.2 N (C) 9.0 N (D) 14.4 N **FM0224**

27. An air tight container having a lid with negligible mass and an area of 8cm^2 is partially evacuated. If a 48N force is required to pull the lid off the container and the atmospheric pressure is 1.0×10^5 Pa, the pressure in the container before it is opened must be

(A) 0.6atm (B) 0.5 atm (C) 0.4 atm (D) 0.2 atm **FM0225**

28. The three vessels shown in fig. have same base area. Equal volumes of a liquid are poured in the three vessels. The force on the base will be



(A) maximum in vessel A (B) maximum in vessel B

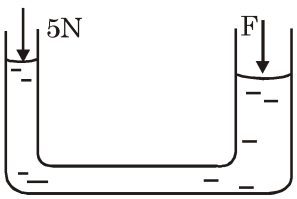
(C) maximum in vssel C (D) equal in all the vessels. **FM0226**

29. A tube of length L is filled completely with incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω. The force exerted by the liquid on the tube at other end is :-

(A) $\frac{M\omega^2 L}{2}$ (B) $M\omega^2 L$ (C) $\frac{M\omega^2 L}{4}$ (D) $\frac{3M\omega^2 L}{4}$

FM0055

30. The area of cross-section of the two arms of a hydraulic press are 1 cm^2 and 10 cm^2 respectively (figure). A force of 5 N is applied on the water in the thinner arm. What force should be applied on the water in the thicker arm so that the water may remain in equilibrium ?

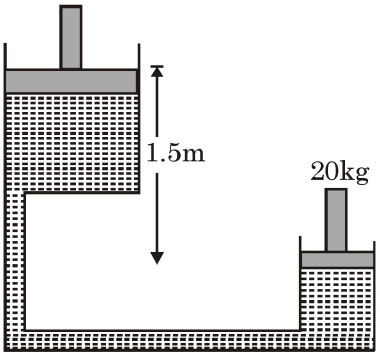


(A) 0.5 N (B) 5 N (C) 50 N (D) 500 N

FM0227

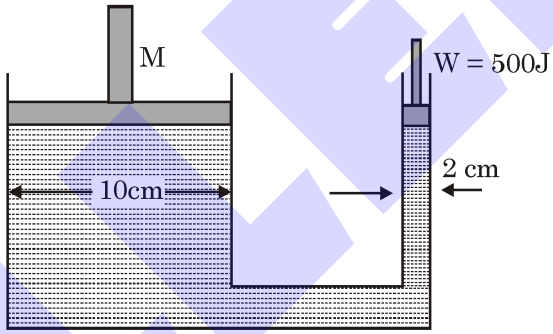
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

31. Figure shows a hydraulic press with the larger piston of diameter 35 cm at a height of 1.5m relative to the smaller piston of diameter 10 cm. The mass on the smaller piston is 20 kg. What is the force exerted on the load by the larger piston. The density of oil in the press is 750 kg/m^3, (Take g = 9.8 m/s^2).



(A) 5×10^3 N (B) 1.3×10^3 N
(C) 3.7×10^3 N (D) 4.8×10^3 N **FM0228**

32. The hydraulic press shown in the figure is used to raise the mass M through a height of 5.0 mm by performing 500 J of work at the small piston. The diameter of the large piston is 10 cm while that of the smaller one is 2 cm. The mass M is



(A) 10^4 kg (B) 10^3 kg (C) 100 kg (D) 10^5 kg

FM0229

Buoyancy

33. Identify the correct choice.
(i) When a body floats in a liquid, it displaces the liquid whose weight is equal to its own weight.
(ii) When a body sinks in a liquid, it displaces the liquid whose volume is equal to its own volume.
(A) (i) is true but (ii) is false. (B) (i) is false but (ii) is true.
(C) Both (i) and (ii) are true. (D) Both (i) and (ii) are false.

FM0230

34. 100 kg of iron and cotton are weighed by using a spring balance on the surface of the Earth.If R_1 and R_2 are the reading shown by the balance,then
(A) $R_1 < R_2$ (B) $R_1 = R_2$
(C) $R_1 > R_2$ (D) $R_1 = R_2 = 0$ **FM0231**

35. A vessel contains oil of density 0.8gm/cc over mercury of density 13.6gm/cc. A sphere floats with half of it's volume immersed in mercury and the other half in the oil. The density of material of sphere. (in gm/cc)
(A) 14.4 (B) 7.2 (C) 3.6 (D) 12.2

FM0232

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\Ad-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

36. A cube of side 20cm is floating on a liquid with 5cm of the cube outside the liquid. If the density of liquid is 0.8 gm/cc then the mass of the cube is

(A) 4.2 kg (B) 4.8 kg (C) 5kg (D) 5.2 kg **FM0233**

37. A cubical box is to be constructed with iron sheets 1 mm in thickness. What can be the minimum value of the external edge so that the cube does not sink in water ? Density of iron = 8000 kg/m^3 and density of water = 1000 kg/m^3.

(A) 32 mm (B) 40 mm (C) 48 mm (D) 52 mm **FM0234**

38. A cubical block of wood weighing 200 g has a lead piece fastened underneath. Find the mass of the lead piece which will just allow the block to float in water. Specific gravity of wood is 0.8 and that of lead is 11.3.

(A) 52.2 gm (B) 54.8 gm (C) 57.0 gm (D) 61.2 gm

FM0235

39. If a block of iron (density 5gm/cm^3) of size 5cm x 5cm x 5cm was weighed while completely submerged in water, what would be the apparent weight? (density of water = 1gm/cm^3)

(A) 5x5x5x5 gm wt (B) 4x4x4x5 gm wt (C) 3x5x5x5 gm wt (D) 4x5x5x5 gm wt

FM0236

40. An ornament weighing 36 g in air, weighs only 34 g in water. Assuming that some copper is mixed with gold to prepare the ornament, find the amount of copper in it. Specific gravity of gold is 19.3 and that of copper is 8.9.

(A) 1.4 gm (B) 1.8 gm (C) 2.1 gm (D) 2.2 gm

FM0237

41. A body of volume 100 c.c. is immersed completely in water contained in a jar. The weight of water and the jar before immersion of the body was 700g wt. After immersion weight of water and jar will be :

(A) 700 g wt (B) 800 g wt (C) 500 g wt (D) 100 g wt

FM0238

42. A sample of metal weights 210 grams in air, 180 grams in water and 120 grams in an unknown liquid. Then:–

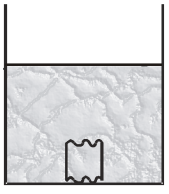
(A) the density of metal is 3 g/cm^3

(B) the density of metal is 7 g/cm^3

(C) density of metal is 4 times the density of the unknown liquid

(D) the metal will float in water **FM0239**

43. A metal piece of mass 160 g lies in equilibrium inside a glass of water (figure). The piece touches the bottom of the glass at a small number of points. If the density of the metal is 8000 kg/m^3, find the normal force exerted by the bottom of the glass on the metal piece.



(A) 1.4 N (B) 2.7 N (C) 3.3 N (D) 3.8 N **FM0240**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

44. A sphere of density d is let fall in a liquid of density d/4. The acceleration of the body will be

(A) $\frac{g}{4}$ (B) $\frac{3g}{4}$ (C) $\frac{g}{2}$ (D) g **FM0241**

45. A hollow metal sphere is found to float in water with the highest point just touching the free surface of water. If 'd' is the density of the metal in cgs units, the fraction that represents the volume of the hollow in terms of the volume of the sphere is

(A) $\frac{1}{d}$ (B) $\left(1-\frac{1}{d}\right)$ (C) $\frac{d}{(d-1)}$ (D) $\left(1+\frac{1}{d}\right)$ **FM0242**

46. A hemispherical bowl just floats without sinking in a liquid of density 1.2×10^3 kg/m^3. If the outer diameter and the density of the bowl are 1m and 2×10^4 kg/m^3 respectively, then the inner diameter of the bowl will be

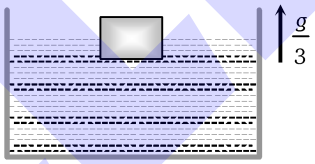
(A) 1.91 m (B) 0.5m (C) 0.98 m (D) 1.75 m **FM0243**

47. A cubical block of wood of edge a and density ρ floats in water of density 2ρ. The lower surface of the cube just touches the free end of a mass less spring of force constant K fixed at the bottom of the vessel. The weight W put over the block so that it is completely immersed in water without wetting the weight is:

(A) $a(a^2\rho g + k)$ (B) $a(a\rho g + 2k)$ (C) $a\left(\frac{a\rho g}{2} + 2k\right)$ (D) $a\left(a^2\rho g + \frac{k}{2}\right)$

FM0244

48. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of *g/3*, the fraction of volume immersed in the liquid will be :



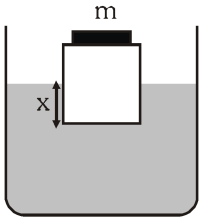
(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ **FM0245**

49. A cork of density $0.5 gcm^{-3}$ floats on a calm swimming pool. The fraction of the cork's volume which is under water is :-

(A) 0% (B) 25% (C) 10% (D) 50%

FM0059

50. A cubical block (a × a × a), with a coin of mass 'm' over it is floating over a liquid of density ρ. In this case x_1 depth of the block is immersed. Now the coin is removed & it is found that x_2 depth is immersed in liquid. Value of $(x_1 - x_2)$ is :-

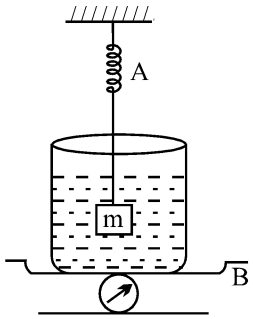


(A) $\frac{m}{\rho a^2}$ (B) $\frac{\rho a^4}{m}$ (C) $\frac{m}{2\rho a^2}$ (D) data insufficient

FM0061

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

51. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in the figure in this situation:

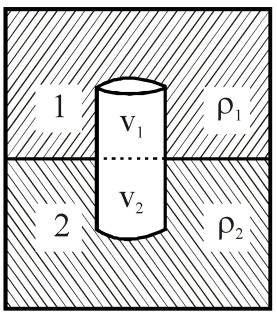


(A) the balance A will read more than 2 kg.
(B) the balance B will read more than 5 kg.
(C) the balance A will read less than 2 kg and B will read less than 5 kg.
(D) the balances A and B will read 2 kg and 5 kg respectively. **FM0062**

52. A solid floats with 2/3 of its volume immersed in a liquid and with 3/4 of its volume immersed in another liquid. What fraction of its volume will be immersed if it floats in a homogenous mixture formed of equal volumes of the liquids?
(A) 6/7 (B) 8/11 (C) 11/16 (D) 12/17 **FM0064**

53. **Statement-1 :** Submarine sailors are advised that they should not allow it to rest on floor of the ocean.
Statement-2 : The force exerted by a liquid on a submerged body may be downwards.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **FM0068**

54. **Statement-1 :** When a body floats such that it's parts are immersed into two immersible liquids then force exerted by liquid-1 is of magnitude $\rho_1 v_1 g$.
Statement-2 : Total Buogyant force = $\rho_1 v_1 g + \rho_2 v_2 g$

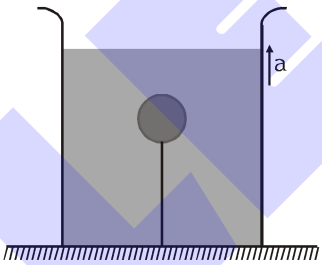


(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **FM0067**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

55. A wooden block floats in water in a sealed container. When the container is at rest, 25% of the block is above the water. Consider the following five situation :
(i) the container is lifted up at constant speed.
(ii) the container is lowered at constant speed.
(iii) the container is lifted up at an increasing speed.
(iv) the container is lowered at a decreasing speed.
(v) the air pressure above the water in the container is increased.
What happens in each situation?
(A) 75% of block is submerged in siutation (i) & (ii) and higher fraction is submerged in situations (iii), (iv) & (v)
(B) 75% of block is submerged in situation (i), (ii) & (v) and higher fraction is submerged in situations (iii), (iv).
(C) 75% of block is submerged in situation (i), (ii), (iii) & (iv) and higher fraction is submerged in situations (v).
(D) 75% of block is submerged in all situation. **FM0122**

56. The tension in a string holding a solid block below the surface of a liquid (of density greater than that of solid) as shown in figure is T_0 when the system is at rest. What will be the tension in the string if the system has an upward acceleration a?



(A) $T_0\left(1+\frac{a}{g}\right)$ (B) $T_0\left(1-\frac{a}{g}\right)$ (C) $2T_0\left(1-\frac{a}{g}\right)$ (D) $3T_0\left(1+\frac{2a}{g}\right)$

FM0121

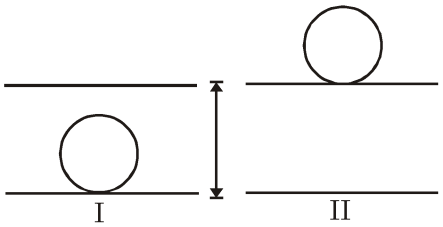
57. A boat full of scrap iron is floating on water in a lake. If all the iron is dropped into the water,the level of water will
(A) go up (B) fall down
(C) remain the same (D) can not be decided **FM0246**

58. In order that a floating object be in a stable equilibrium, its centre of bouyancy should be
(A) vertical below its centre of gravity
(B) horizontally inline with its centre of gravity
(C) vertically above its centre of gravity
(D) may be anywhere **FM0247**

59. The densities of wood and benzene at 0°C are 880 kg/m^3 and 900 kg/m^3 respectively. The coefficients of volume expansion are 1.2×10^{-3}/°C for wood and 1.5×10^{-3}/°C for benzene. At what temperature will a piece of wood just sink in benzene?
(A) 47°C (B) – 47°C (C) 83°C (D) –83°C **FM0248**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

60. In figure-I is shown a sphere of mass m and radius r resting at the bottom of a large container filled with water. Depth of the container is h. Density of material of the sphere is the same as that of water. Now the whole sphere is slowly pulled out of water as shown in figure-II



Work done by the agent in pulling the sphere is equal to

(A) mgr (B) 0.5mgr (C) mg(0.5r+h) (D) mg(r+h) **FM0249**

61. A rectangular block of wood of density 800kgm^{-3} having a mass of 2kg is pushed in to water so that it is completely submerged and then released. Neglecting viscous forces, the initial acceleration of the block will be (g = 10 m/s^2)

(A) 1.25 m/s^2 downward (B) 2.5 m/s^2 upward

(C) 1.25 m/s^2 upward (D) 2.5 m/s^2 downward **FM0250**

Surface Tension

62. Two merucry drops (each of radius 'r') merge to form a bigger drop. The surface energy of the bigger drop, if $\frac{1}{\pi}$ is the surface tension (in SI unit), is :

(A) $2^{5/3}r^2$ (B) $4r^2$ (C) $2r^2$ (D) $2^{8/3}r^2$

FM0082

63. By a surface of a liquid we mean

(A) a geometrical plane like x = 0

(B) all molecules exposed to the atmosphere

(C) a layer of thickness of the order of 10^{-8} m

(D) a layer of thickness of the order of 10^{-4} m **FM0251**

64. An ice tube is suspended in vacuum in a gravity-free hall. As the ice melts it

(A) will retain its cubical shape

(B) will change its cubical shape

(C) will fall down on the floor of the hall

(D) will fly up. **FM0252**

65. Find the surface energy of water kept in a cylindrical vessel of radius 6.0 cm. Surface tension of water = 0.075 J/m^2.

(A) 4.25×10^{-4} J (B) 8.5×10^{-4} J (C) 12.75×10^{-4} J (D) 1.7×10^{-3} J

FM0253

66. Work of 6.0×10^{-4} joule is required to be done in increasing the size of a soap film from 10 cm $\times$ 6 cm to 10 cm $\times$ 11cm. The surface tension of the film is (in N/m)

(A) 5×10^{-2} (B) 6×10^{-2} (C) 1.5×10^{-2} (D) 1.2×10^{-2}

FM0254

67. A film of water is formed between two straight parallel wires of length 10cm each separated by 0.5 cm. If their separation is increased by 1 mm while still maintaining their parallelism how much work will have to be done (Surface tension of water = 7.2×10^{-2}N /m)

(A) 7.22×10^{-6} J (B) 1.44×10^{-5} J (C) 2.88×10^{-5} J (D) 5.76×10^{-5} J

FM0255

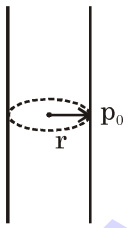
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

68. The work done is blowing a soap bubble of volume "V" is W. The work done in blowing a soap bubble of volume "2V" is

(A) W (B) $2^{\frac{2}{3}}W$ (C) $3^{\frac{2}{3}}W$ (D) 2W **FM0256**

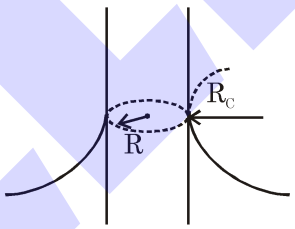
69. If two soap bubbles of different radii are connected by a tube,
(A) air flows from the bigger bubble to the smaller bubble till the sizes become equal
(B) air flows from bigger bubble to the smaller bubble till the sizes are interchanged
(C) air flows from the smaller bubble to the bigger
(D) there is no flow of air. **FM0079**

70. A cylindrical stream of water is falling in air. If radius of cylinder is r, atmospheric pressure is p_0 and surface tension is S, pressure inside cylinder is



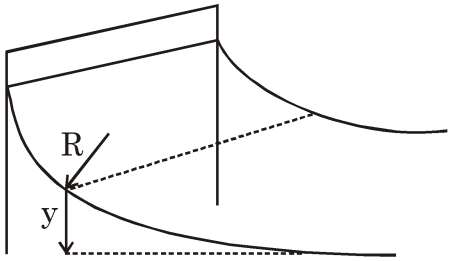
(A) $p_0 + \frac{2s}{r}$ (B) $p_0 + \frac{4s}{r}$ (C) $p_0 + \frac{s}{r}$ (D) $p_0 + \frac{3s}{r}$ **FM0257**

71. A glass rod of radius R is dipped in water which completely wets the glass. Water rises as shown. The space is gravity free. The radius of curvature of the free surface R_C as shown.



(A) R (B) $\frac{R}{2}$ (C) 2R (D) 4R **FM0258**

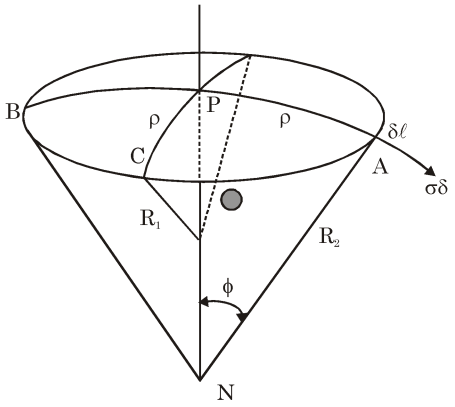
72. Near a wall, we fill a completely wetling liquid. If surface tension of liquid is S and its density ρ, what is the radius of curvature at a height y from Horizontal free surface.



(A) $\frac{2s}{y\rho g}$ (B) $\frac{4s}{y\rho g}$ (C) $\frac{s}{y\rho g}$ (D) $\frac{s}{2y\rho g}$ **FM0259**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

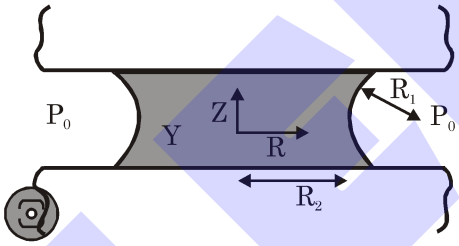
73. If a liquid drop has radii of $R_1 = 1m$ & $R_2 = 2m$ in two mutually perpendicular directions as shown, the excess pressure inside the drop is _____N/m^2. Take S = 0.07 N/m.



(A) 0.07 N/m^2 (B) 0.105 N/m^2 (C) 0.035 N/m^2 (D) None of these

FM0260

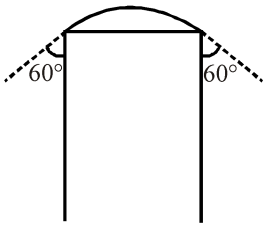
74. If a liquid drop has radii of $R_1 = 1m$ & $R_2 = 2m$ in two mutually perpendicular directions the excess pressure inside the drop is _____N/m^2. Take S = 0.07 N/m.



(A) 0.035 N/m^2 (B) – 0.035 N/m^2 (C) 0.105 N/m^2 (D) 0.07 N/m^2

FM0261

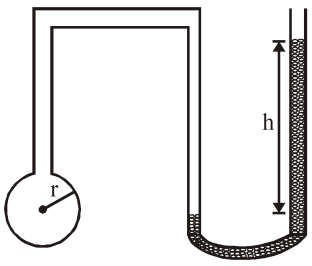
75. A soap bubble is being blown on a tube of radius 1 cm. The surface tension of the soap solution is 0.05 N/m and the bubble makes an angle of 60° with the tube as shown. The excess of pressure over the atmospheric pressure in the tube is :



(A) 5 Pa
(B) 1 Pa
(C) 10 Pa
(D) 20 Pa

FM0080

76. If the radius of the Soap-bubble on one side of tube is r and difference in height of liquid of density ρ in manometer is h, then surface tension of liquid used to make the bubble is :-



(A) $T = 2r\rho hg$ (B) $T = \frac{rh\rho g}{4}$ (C) $T = \frac{2\pi rh\rho g}{2}$ (D) $T = \frac{rh\rho g}{2}$

FM0113

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

77. There is an air bubble of radius 1.0 mm in a liquid of surface tension 0.075 N/m and density 1000 kg/m^3. The bubble is at a depth of 10 cm below the free surface. By what amount is the pressure inside the bubble greater than the atmospheric pressure ? Take g = 9.8 m/s^2.

(A) 890 Pa (B) 952 Pa (C) 1130 Pa (D) 1208 Pa **FM0262**

78. The pressure inside soap bubble is 1.01 and 1.02 atmosphere respectively the ratio of their volume

(A) 102:101 (B) $(102)^3 : (101)^3$ (C) 8 : 1 (D) 2 : 1 **FM0263**

79. Two soap bubbles of radii 3mm and 4mm are in contact radius of curvature of interface between those two bubbles is

(A) 1mm (B) 7mm (C) 12mm (D) $\frac{12}{7}$mm **FM0264**

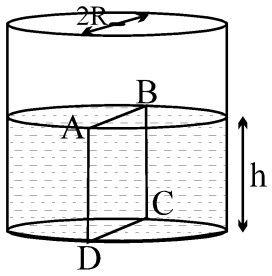
80. A spherical soap bubble of radius 1cm is formed inside another of radius 3cm the radius of single soap bubble which maintains the same pressure difference as inside the smaller and outside the larger soap bubble is ___ cm

(A) 4/3 (B) 3/4 (C) 1/2 (D) 2 **FM0265**

81. An air bubble of diameter, d = 4μm is located in water at a depth h = 5.0m Considering standard atmospheric prssure as 1 atm, find the pressure in the air - bubble? (S = 0.07 N/m)

(A) 2.2 atm (B) 1.2 atm (C) 3.2 atm (D) 1.6 atm **FM0266**

82. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude **[IIT-JEE 2007]**



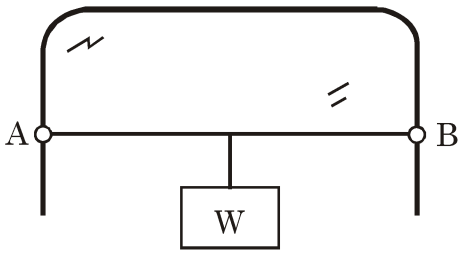
(A) $|2P_0Rh + \pi R^2\rho gh - 2RT|$
(B) $|2P_0Rh + R\rho gh^2 - 2RT|$
(C) $|P_0\pi R^2 + R\rho gh^2 - 2RT|$
(D) $|P_0\pi R^2 + R\rho gh^2 + 2RT|$

FM0089

83. A metallic wire of diameter "d" is lying horizontally on the surface of water. The maximum length of wire so that is may not sink will be (T is surface tension)

(A) $\sqrt{\frac{2T}{\pi dg}}$ (B) $\sqrt{\frac{2Tg}{\pi d}}$ (C) $\sqrt{\frac{2\pi d}{Tg}}$ (D) any length **FM0267**

84. A light wire AB of length 10 cm can slide on a vertical frame as shown in figure. There is a film of soap solution trapped between the frame and the wire. Find the load W that should be suspended from the wire to keep it in equilibrium. Neglect friction. Surface tension of soap solution = 25 dyne/cm. Take g = 10 m/s^2.



(A) 5×10^{-3} N (B) 2.5×10^{-3} N (C) 5×10^{-4} N (D) 10×10^{-3} N

FM0268

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

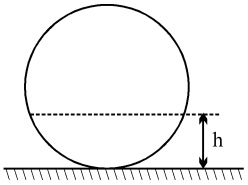
85. A 5.0 cm long straight piece of thread is kept on the surface of water. Find the force with which the surface on one side of the thread pulls it. Surface tension of water = 0.076 N/m.

(A) 1.9 mN (B) 2.7 mN (C) 3.4 mN (D) 3.8 mN **FM0269**

86. A glass plate of length 20 cm and breadth 0.2 cm just touches the water surface in a beaker. The Surface tension of water is 72 dyne/ cm. The weight of the glass plate is 25g. The weight that must be placed in the right pan to counter pose the balance is

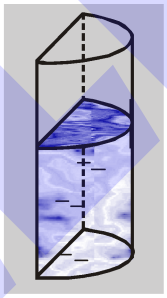
(A) 25 g (B) 28 g (C) 22 g (D) 21.3g **FM0270**

87. A liquid is filled in a spherical container of radius R till a height h. At this position the liquid surface at the edges is also horizontal. The contact angle is :-



(A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$ (C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$ **FM0083**

88. A liquid is contained in a vertical tube of semicircular cross-section fig. The contact angle is zero. The force of surface tension on the curved part and on the flat part are in ratio



(A) 1 : 1 (B) 1 : 2 (C) π : 2 (D) 2 : π **FM0271**

89. In the above situation , if contact angle is 90°, what is the ratio :-

(A) 1 : 1 (B) 1 : 2 (C) π : 2 (D) 2 : π **FM0272**

90. If surface tension of liquid air boundary is $S_{\ell a}$, surface tension of solid air surface is S_{Sa} and surface tension of liquid solid boundary is $S_{\ell S}$, the contact angle θ is given by

(A) $\cos\theta = \frac{S_{S\ell} - S_{\ell a}}{S_{Sa}}$ (B) $\cos\theta = \frac{S_{Sa} - S_{\ell a}}{S_{S\ell}}$

(C) $\cos\theta = \frac{S_{Sa} - S_{S\ell}}{S_{\ell a}}$ (D) None of these **FM0273**

91. In which of the following cases is the liquid said to completely wet the surface. Symbols have usual notation.

(A) $S_{\ell a} \geq S_{Sa} + S_{S\ell}$ (B) $S_{Sa} \geq S_{S\ell} + S_{\ell a}$

(C) $S_{S\ell} \geq S_{Sa} + S_{\ell a}$ (D) None of these **FM0274**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

92. The angle of contact between glass and water is 0° and water (surface tension 70 dyn/cm) rises in a glass capillary up to 6 cm. Another liquid of surface tension 140dyn/cm. Angle of contact 60° and relative density 2 will rise in the same capillary up to

(A) 12 cm (B) 24 cm (C) 3cm (D) 6cm **FM0275**

93. The lower end of a capillary tube is at a depth of 12cm and water rises 3cm in it. Assume contact angle is 0°. The mouth pressure required to blow an air bubble at the lower end will be x cm of water column, where x is :

(A) 12 (B) 15 (C) 3 (D) 9 **FM0276**

94. A vertical glass capillary with inside diameter 0.50 mm is submerged into water so that the length of its part emerging outside the water surface is equal to 25 mm.Find the radius of curvature of the meniscus.Surface tension of water is 73×10^{-3} N/m .

(A) R = 0.6m (B) R = 6mm

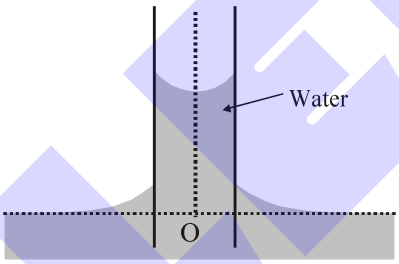
(C) R = 0.6mm (D) R = 0.6Km **FM0277**

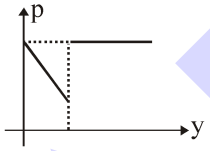
95. If 'L' is the capillary rise or dip and 'A' the cross sectional area of the tube, other conditions being the same, then

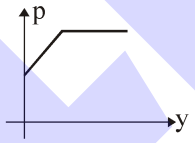
(A) LA = Constant (B) $L\sqrt{A}$ = Constant

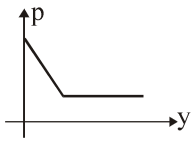
(C) L/A = Constant (D) $L/\sqrt{A}$ = Constant **FM0278**

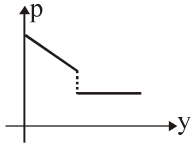
96. Water rises in a capillary as shown. The correct graph of pressure (p) vs height (y) above point O along the axis of capillary is :-



(A) 

(B) 

(C) 

(D) 

FM0086

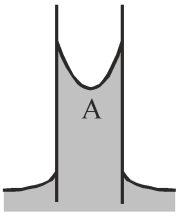
97. A 20 cm long capillary tube is dipped in water. The water rises upto 8 cm. If the entire arrangement is put in a freely falling elevator, the length of water column in the capillary tube will be-

[AIEEE - 2005]

(A) 8 cm (B) 10 cm (C) 4 cm (D) 20 cm

FM0088

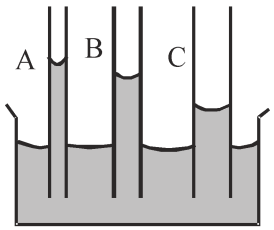
98. Fig. shows a capillary tube of radius r dipped into water. If the atmospheric pressure is P_0, the pressure at point A is



(A) P_0 (B) $P_0 + \frac{2S}{r}$ (C) $P_0 - \frac{2S}{r}$ (D) $P_0 - \frac{4S}{r}$ **FM0279**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

99. The capillaries in figure have inner radii 0.5 mm, 1.0 mm and 1.5 mm respectively. The liquid in the beaker is water. Find the sum of heights rises of water level in the three capillaries. The surface tension of water is 7.5×10^{-2} N/m.



(A) 5.5 cm (B) 5 cm (C) 6 cm (D) 7.5 cm

FM0280

100. A capillary tube of radius 0.50 mm is dipped vertically in a pot of water. Find the difference between the pressure of the water in the tube 5.0 cm below the surface and the atmospheric pressure. Surface tension of water = 0.075 N/m.

(A) 100 Pa (B) 150 Pa (C) 160 Pa (D) 190 Pa **FM0281**

101. A capillary tube is taken from the Earth to the surface of the Moon. The rise of the liquid column on the Moon (acceleration due to gravity on the Earth is 6 times that of the Moon) is

(A) six times that on the Earth surface. (B) $\frac{1}{6}$ that on the Earth's surface.

(C) equal to that on the Earth's surface. (D) zero **FM0282**

102. When a capillary tube is lowered into water, the mass of the water raised above the outside level is 5 gm. If the radius of the tube is doubled the mass of water that raises in the tube above the outside level is:

(A) 1.25 gm (B) 5 gm (C) 10 gm (D) 20 gm **FM0283**

103. A vessel has a small hole at its bottom. If water can be poured into it upto a height of 7 cm without leakage ($g = 10\ ms^{-2}$), the radius of the hole is (surface tension of water is $0.07\ Nm^{-1}$)

(A) 2 mm (B) 0.2 mm (C) 0.1 mm (D) 0.4 mm

FM0284

104. An open capillary tube is lowered in a vessel with mercury. The difference between the levels of the mercury in the vessel and in the capillary tube $\Delta h = 4.6$mm. What is the radius of curvature of the mercury meniscus in the capillary tube? Surface tension of mercury is 0.46 N/m, density of mercury is 13.6 gm/cc.

(A) $\frac{1}{340}$ m (B) $\frac{1}{680}$ m (C) $\frac{1}{1020}$ m (D) Information insufficient

FM0084

105. A glass capillary tube of inner diameter 0.28 mm is lowered vertically into water in a vessel. The pressure to be applied on the water in the capillary tube so that water level in the tube is same as the vessel in N/m^2 is (surface tension of water = 0.07 N/m atmospheric pressure = $10^5\ N/m^2$)

(A) 10^3 (B) 99×10^3 (C) 100×10^3 (D) 101×10^3

FM0285

106. The lower end of a capillary tube of radius r is placed vertically in water then with the rise of water in the capillary heat evolved is

(A) $+\frac{\pi r^2 h^2 dg}{2J}$ (B) $+\frac{\pi r^2 h^2 dg}{J}$ (C) $-\frac{\pi r^2 h^2 dg}{2J}$ (D) $-\frac{\pi r^2 h^2 dg}{J}$

FM0286

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

107. A glass rod of radius r_1 is inserted symmetrically into a vertical capillary tube of radius r_2 such that their lower ends are at same level. The arrangement is now dipped in water the height to which water will enter into the tube (σ = surface tension, ρ = density of water, $\theta = 0°$, r_2 is only slightly more than r_1)

(A) $\frac{2\sigma}{(r_2 - r_1)\rho g}$ (B) $\frac{\sigma}{(r_2 - r_1)\rho g}$ (C) $\frac{2\sigma}{(r_2 + r_1)\rho g}$ (D) $\frac{2\sigma}{(r_2^2 + r_1^2)\rho g}$

FM0287

108. A long capillary tube of radius 1 mm, open at both ends is filled with water and placed vertically. What will be the height of water column left in the capillary ? (Surface tension of water is 73.5×10^{-3} Nm^{-1}, $\theta = 0°$)

(A) 0.3 cm (B) 3cm (C) 6cm (D) 0.03 cm

FM0288

109. The surface tension of a liquid __with rise of temperature.

(A) increases (B) decreases

(C) remains same (D) first decreased and then increases

FM0289

110. At critical temperature surface tension becomes

(A) 0 (B) 1 (C) infinite (D) negative **FM0290**

Fluid dynamics

111. Two water pipes P and Q having diameters 2×10^{-2} m and 4×10^{-2}m, respectively, are joined in series with the main supply line of water. The velocity of water flowing in pipe P is :-

(A) 4 times that of Q (B) 2 times that of Q

(C) 1/2 times of that of Q (D) 1/4 times that of Q **FM0071**

112. At Deoprayag (Garhwal, UP) river Alaknanda mixes with the river Bhagirathi and becomes river Ganga. Suppose Alaknanda has a width of 12 m, bhagirathi has a width 8 m and Ganga has a width of 16 m. Assume that the depth of water is same in the three rivers. Let the average speed of water in Alaknanda be 20 km/h and in Bhagirathi be 16 km/h. Find the average speed of water in the river Ganga.

(A) 23 km/h (B) 46 km/h (C) 20 km/h (D) 40 km/h

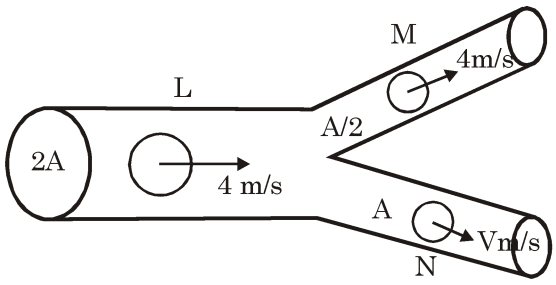
FM0291

113. Water flows through a non-uniform tube of area of cross sections A, B and C whose values are 25, 15 and 35 cm^2 respectively. The ratio of the velocities of water at the sections A, B and C is

(A) 5 : 3 : 7 (B) 7 : 3 : 5 (C) 21 : 35 : 15 (D) 1 : 1 : 1

FM0292

114. An incompressible liquid flows through a horizontal tube L M N as shown in the figure. Then the velocity 'V' of the liquid through the tube N is :



(A) 1 ms^{-1} (B) 2 ms^{-1} (C) 4.5 ms^{-1} (D) 6 ms^{-1} **FM0293**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.pt65

115. **Statement–1 :** The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.
and
Statement–2 : In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant. **[IIT-JEE 2008]**
(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1
(C) Statement–1 is True, Statement–2 is False
(D) Statement–1 is False, Statement–2 is True **FM0294**

116. In the case of a fluid, Bernoulli's theorem expresses the application of the principle of conservation of :-
(A) linear momentum (B) energy
(C) mass (D) angular momentum **FM0070**

117. Water is flowing steadily through a horizontal tube of non uniform cross-section. If the pressure of water is 4×10^4 N/m^2 at a point where cross-section is 0.02 m^2 and velocity of flow is 2 m/s, what is pressure at a point where cross-section reduces to 0.01 m^2.
(A) 1.4×10^4 N/m^2 (B) 3.4×10^4 N/m^2
(C) 2.4×10^{-4} N/m^2 (D) none of these **FM0069**

118. Water is flowing in a pipe of uniform cross section under constant pressure difference. At some place the pipe becomes narrow. The pressure of water at this place
(A) remains same (B) may increase or decrease
(C) increases (D) decreases **FM0295**

119. If air is blown through the space between a calendar suspended from a nail on wall and the wall, then
(A) the calendar moves close to the wall.
(B) the calendar moves farther from the wall.
(C) the position of the calendar does not change.
(D) the position of the calendar may or may not change. **FM0296**

120. The velocity of the wind over the surface of the wing of an aeroplane is 80 ms^{-1} and under the wing 60 ms^{-1}. If the area of the wing is 4m^2, the dynamic lift experienced by the wing is:
[Density of air = 1.3 kg. m^{-3}]
(A) 3640 N (B) 7280 N (C) 14560 N (D)72800 N
FM0297

121. At the mouth of the tap, the area of cross-section is 2.0 cm^2 and the speed of water is 3m/s. The area of cross-section of the water column 80cm below the tap is(use g = 10 m/s^2)
(A) 0.6 cm^2 (B) 1.2 cm^2 (C) 1.5 cm^2 (D) 2.0 cm^2
FM0298

122. In a horizontal pipe line of uniform cross-section, pressure falls by 5Pa between two points separated by 1km. The change in the kinetic energy per kg of the oil flowing at these points is:
(Density of oil = 800 kgm^3)
(A) 6.25×10^{-3} Jkg^{-1} (B) 5.25×10^{-4} Jkg^{-1}
(C) 3.25×10^{-5} Jkg^{-1} (D) 4.25×10^{-2} Jkg^{-1} **FM0299**

123. At a point P in a water pipe line the velocity is 1 ms^{-1} and the pressure is 3×10^5 pa . At another point Q the area of cross section is half that of at P and the pressure is 5×10^5 pa . The difference of heights between P and Q in metre is (g = 10 ms^{-2})
(A) 10.5 (B) 20.15 (C) 4.5 (D) zero **FM0300**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

124. A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom, is- **[AIEEE - 2002]**

(A) 10 (B) 20 (C) 25.5 (D) 5

FM0072

125. Water is filled in a container upto height 3m. A small hole of area 'a' is punched in the wall of the container at a height 52.5 cm from the bottom. The cross sectional area of the container is A. If $\frac{a}{A}=0.1$ then v^2 is (where v is the velocity of water coming out of the hole) **[IIT-JEE' 2005 (Scr)]**

(A) 48 (B) 51 (C) 50 (D) 51.5

FM0078

126. Two identical tall jars are filled with water to the brim. The first jar has a small hole on the side wall at a depth h/3 and the second jar has a small hole on the side wall at a depth of 2h/3, where 'h' is the height of the jar. The water issuing out from the first jar falls at a distance R_1 from the base and the water issuing out from the second jar falls at a distance R_2 from the base. The correct relation between R_1 and R_2 is

(A) $R_1 > R_2$ (B) $R_1 < R_2$ (C) $R_2 = 2R_1$ (D) $R_1 = R_2$ **FM0301**

127. There are two holes one each along the opposite sides of a wide rectangular tank. The cross section of each hole is $0.01m^2$ and the vertical distance between the holes is one meter. The tank is filled with water. The net force on the tank in newton when water flows out of the holes is: (Density of water $1000kg/m^3$)

(A) 100 (B) 200 (C) 300 (D) 400 **FM0302**

128. A tank with vertical walls is mounted so that its base is at a height H above the horizontal ground. The tank is filled with water to a depth 'h' . A hole is punched in the side wall of the tank at a depth ' x ' below the water surface. To have maximum range of the emerging stream, the value of x is

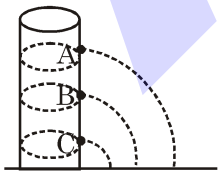
(A) $\frac{H+h}{4}$ (B) $\frac{H+h}{2}$ (C) $\frac{H+h}{3}$ (D) $\frac{3(H+h)}{4}$ **FM0303**

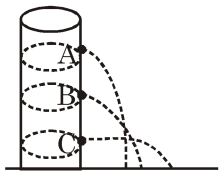
129. A hole is made at the bottom of a tank filled with water (density = $10^3 k\ g/m^3$). If the total pressure at the bottom of the tank is 3 atm (1 atm = $10^5\ N/m^2$) , then the velocity of efflux is

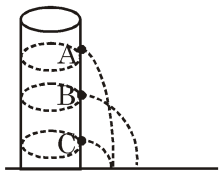
(A) $\sqrt{400}$ m/s (B) $\sqrt{200}$ m/s (C) $\sqrt{600}$ m / s (D) $\sqrt{500}$ m / s

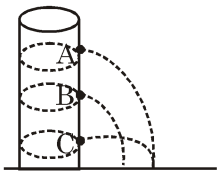
FM0304

130. Water stands at a height of 100 cm in a vessel whose side walls are vertical. A , B and C are holes at height 80cm, 50 cm, and 20 cm respectively from the bottom of the vessel. The correct system of water flowing out is :

(A) 

(B) 

(C) 

(D) 

FM0305

131. Tanks A and B open at the top contain two different liquids upto certain height in them. A hole is made on the wall of each tank at a depth 'h' from the surface of the liquid. The area of the hole in 'A' is twice that of in B. If the liquid mass flux through each hole is equal, then the ratio of the densities of the liquids respectively is

(A) $\frac{2}{1}$ (B) $\frac{3}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$ **FM0306**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

132. A cylindrical tank has a hole of 2 cm^2 at its bottom. If the water is allowed to flow into the tank from a tube above it at the rate of 100 cm^3/s, then find the maximum height upto which water can rise in the tank (Take g = 10 m s^{-2})

(A) 2.5×10^{-2} m (B) 1.25×10^{-2} m

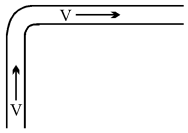
(C) 5.5×10^{-2} m (D) 3.5×10^{-2} m **FM0307**

133. A tank of height 5 m is full of water. There is a hole of cross sectional area 1 cm^2 in its bottom. The initial volume of water that will come out from this hole per second is :–

(A) $10^{-3}\,m^3/s$ (B) $10^{-4}\,m^3/s$ (C) $10\,m^3/s$ (D) $10^{-2}\,m^3/s$

FM0308

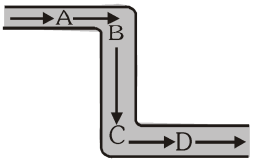
134. A fire hydrant delivers water of density ρ at a volume rate L. The water travels vertically upward through the hydrant and then does 90° turn to emerge horizontally at speed V. The pipe and nozzle have uniform cross-section throughout. The force exerted by the water on the corner of the hydrant is :-



(A) ρVL (B) zero (C) 2ρVL (D) $\sqrt{2}\rho VL$

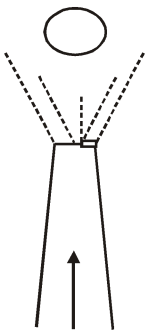
FM0076

135. The tube shown is of non-uniform cross-section. The cross-section area at A is half of the cross-section area at B,C and D. A liquid is flowing through in steady state.The liquid exerts on the tube-



Statement I : A net force towards right.
Statement II : A net force towards left.
Statement III : A net force in some oblique direction.
Statement IV : Zero net force
Statement V: A net clockwise torque.
Statement VI: A net counter-clockwise torque.
Out of these
(A) Only statement I and V are correct
(B) Only statement II and VI are correct
(C) Only statement IV and VI are correct
(D) Only statement III and VI are correct **FM0309**

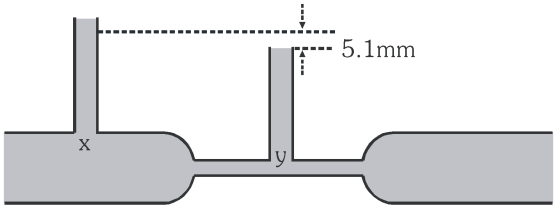
136. **Statement-1:** A light celluloid ball placed in a stream of gas or water issuing at a high velocity from a tube with a narrow neck, the ball floats freely in this stream (Fig)



Statement -2: The gas in the stream has a high velocity, the pressure inside the stream is above atmospheric.
(A) Statement 1 is true, statement-2 true and statement-2 is a correct explanation for statement-1
(B) Statement 1 is true, statement 2 is true, statement-2 is not a correct explanation for statement 1
(C) Statement 1 is true; statement 2 is false
(D) Statement 1 is false; statement 2 is true. **FM0310**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

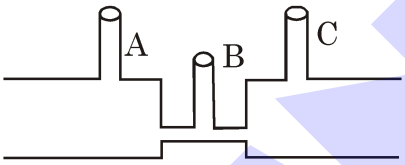
137. The diagram (figure) shows a venturimeter, through which water is flowing. The speed of water at X is 2 cm/s. The speed of water at Y (taking g = 1000 cm/s^2) is :–



(A) 23 cm/s (B) 32 cm/s (C) 101 cm/s (D) 1024 cm/s

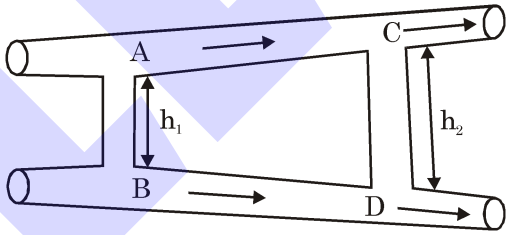
FM0311

138. Three tubes A, B, C are connected to a horizontal pipe in which liquid is flowing. The radii of the pipes at the joints of A, B and C are 2 cm, 1 cm and 2 cm respectively. The height of the liquid



(A) in A is lesser than C (B) in A and C is equal
(C) is same in all the three (D) in A and B is same **FM0312**

139. An ideal liquid is flowing in two pipes, AC is inclined and BD is horizontal. Both the pipes are connected by two vertical tubes of length h_1 and h_2 as shown in the Fig. The flow is streamlined in both the pipes. Assume that no fluid flows through AB & CD. If velocity of liquid at A, B and C are 2m/s, 4m/s and 4m/s respectively, the velocity at D will be

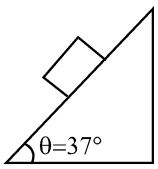


(A) 4 m/s (B) $\sqrt{14}$ m/s (C) $\sqrt{28}$ m/s (D) None

FM0313

Viscosity

140. A cubical block of side 'a' and density 'ρ' slides over a fixed inclined plane with constant velocity 'v'. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be:



(A) $\frac{3\rho a g t}{5v}$ (B) $\frac{4\rho a g t}{5v}$ (C) $\frac{\rho a g t}{v}$ (D) none of these

FM0092

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

141. There is a 1mm thick layer of glycerine between a flat plate of area 100 cm^2 & a big fixed plate. If the coefficient of viscosity of glycerine is 1.0 kg/m-s then how much force is required to move the plate with a velocity of 7 cm/s?
(A) 3.5 N (B) 0.7 N (C) 1.4 N (D) None
FM0093

142. A large wooden plate of area 10 m^2 floating on the surface of a river is made to move horizontally with a speed of 2 m/s by applying a tangential force. If the river is 1 m deep and the water in contact with the bed is stationary, find the tangential force needed to keep the plate moving. Coefficient of viscosity of water at the temperature of the river = 10^{-2} poise.
(A) 15 mN (B) 18 mN (C) 20 mN (D) 23 mN **FM0314**

143. If the shearing stress between the horizontal layers of water in a river is 1.5 milli newton/ m^2 and $\eta_{water} = 1 \times 10^{-3}$ Pa.s, The velocity gradient is .....s^{-1}.
(A) 1.5 (B) 3 (C) 0.7 (D) 1 **FM0315**

144. A force of 10N is required to draw rectangular glass plate on the surface of a liquid with some velocity. Force needed to draw another glass plate of 3 times length and 2 times width is
(A) 5/3N (B) 10N (C) 60N (D) 30N **FM0316**

145. Water is flowing in a river. If the velocity of a layer at a distance of 10 cm from the bottom is 20 cm/s. Find the velocity of layer at a height of 40 cm from the bottom
(A) 10 cm/s (B) 20 cm/s (C) 30 cm/s (D) 80 cm/s **FM0317**

146. Two drops of same radius are falling through air with steady velocity of v cm/s. If the two drops coalesce, what would be the terminal velocity?
(A) 4 v (B) $(4)^{1/3}$ v (C) 2 v (D) 64 v **FM0090**

147. Spherical balls of radius R are falling in a viscous fluid of viscosity η with a velocity v. the retarding viscous force acting on the spherical ball is- **[AIEEE - 2004]**
(A) directly proportional to R but inversely proportional to v
(B) directly proportional to both radius R and velocity v
(C) inversely proportional to both radius R and velocity v
(D) inversely proportional to R but directly proportional to velocity v **FM0091**

148. An air bubble of radius 1cm is found to rise in a cylindrical vessel of large radius at a steady rate of 0.2 cm per second. If the density of the liquid is 1470 kg m^{-3}, then coefficient of viscosity of liquid is approximately equal to:
(A) 1630 poise (B) 163 centi-poise
(C) 140 poise (D) 140 centi-poise **FM0095**

149. If the terminal speed of a sphere of gold (density = 19.5 g/cm^3) is 0.2 m/s in a viscous liquid (density = 1.5 g/cm^3), find the terminal speed of a sphere of silver (density=10.5 g/cm^3) of the same size in the same liquid. **[AIEEE- 2006]**
(A) 0.4 m/s (B) 0.133 m/s (C) 0.1 m/s (D) 0.2 m/s
FM0097

150. A spherical ball falls through viscous medium with terminal velocity v. If this ball is replaced by another ball of the same mass but half the radius, then the terminal velocity will be (neglect the effect of buoyancy.)
(A) v (B) 2 v (C) 4 v (D) 8v **FM0318**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

151. A solid metallic sphere of radius r is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?

(A) r^2 (B) r (C) $r^{3/2}$ (D) $r^{1/2}$

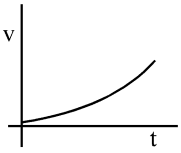
FM0129

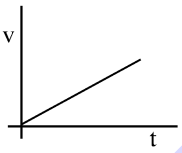
152. The velocity of a ball of mass 'm' density 'd_1' when dropped in a container filled with glycerin of density 'd_2' becomes constant after sometime. The viscous force acting on the ball will be

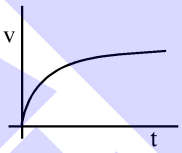
(A) $mg\left(\frac{d_1}{d_2}\right)$ (B) $mg\left(1-\frac{d_2}{d_1}\right)$

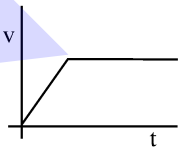
(C) $mg\left(\frac{d_1+d_2}{d_1}\right)$ (D) $mg\left(\frac{d_1+d_2}{d_2}\right)$ **FM0319**

153. Which of the following graphs best represents the motion of a raindrop?

(A) 

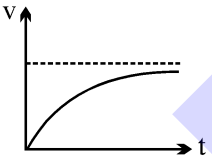
(B) 

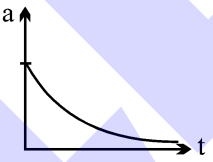
(C) 

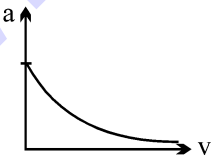
(D) 

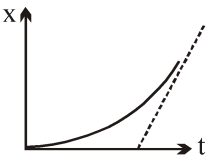
FM0130

154. Which of the following is the ***incorrect*** graph for a sphere falling in a viscous liquid? (Given at t = 0, velocity v = 0 and displacement x = 0.)

(A) 

(B) 

(C) 

(D) 

FM0131

155. Two identical lead shots are dropped at the same time in two glass jars containing water and glycerin. The lead shot dropped in glycerin descends slowly because

(A) viscous force is more in water than in glycerin

(B) viscous force is more in glycerin than in water

(C) surface tension is more in water

(D) surface tension is more in glycerin **FM0320**

156. Water flows at a speed of 6 cm s^{-1} through a tube of radius 1 cm. coefficient of viscosity of water at room temperature is 0.01 poise. the Reynolds number is:

(A) 1000 (B) 120 (C) 1200 (D) 1400 **FM0321**

157. The liquid flow is most stream lined when

(A) Liquid of high viscosity and high density flowing through a tube of small radius.

(B) Liquid of high viscosity and low density flowing through a tube of small radius

(C) Liquid of low viscosity and low density flowing through a tube of large radius

(D) Liquid of low viscosity and high density flowing through a tube of large radius **FM0322**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

158. If the flow is stream lined then Reynolds number is less than
(A) 2000 (B) 3000 (C) 1000 (D) 4000 **FM0323**

159. When temperature is increased:
(a) viscosity of the gas increases
(b) viscosity of the gas decreases
(c) viscosity of the liquid decreases
(d) viscosity of the liquid increases
(A) a and c are true
(B) b and c are true
(C) b and d are true
(D) a and d are true **FM0324**

160. A piece of wood is taken deep inside a long column of water and released. It will move up
(A) with a constant upward acceleration.
(B) with a decreasing upward acceleration.
(C) with a deceleration.
(D) with a uniform velocity. **FM0325**

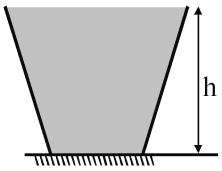
161. Hot syrup flows faster because
(A) surface tension increases with temperature.
(B) viscosity decreases with temperature.
(C) viscosity increases with temperature.
(D) surface tension decreases with temperature. **FM0326**

162. The flow rate from a tap of diameter 1.25 cm is 3 litre/min. The coefficient of viscosity of water is 10^{-2} Pa s. Characterise the flow
(A) stream line (B) turbulent (C) A and B (D) none **FM0327**

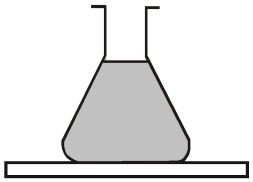
MULTIPLE CORRECT TYPE QUESTIONS

163. A beaker is filled in with water is accelerated a m/s^2 in +x direction. The surface of water shall make an angle:
(A) $\tan^{-1}(a/g)$ backwards
(B) $\tan^{-1}(a/g)$ forwards
(C) $\cot^{-1}(g/a)$ backwards
(D) $\cot^{-1}(g/a)$ forwards **FM0100**

164. M gm of a liquid of density ρ is filled in a light beaker and kept on a horizontal table as shown in the figure. The height of the liquid in the beaker is h. The beaker is wider on top than at its base and the cross-sectional area of the base is A. Neglect the effect of atmospheric pressure. Now, choose the **CORRECT** statement(s) from the following.
(A) The pressure of liquid at the bottom surface is ρgh.
(B) The normal reaction exerted by the table on the beaker is ρghA.
(C) The pressure of the liquid at the bottom surface is $\frac{Mg}{A}$.
(D) The normal reaction exerted by the table on the beaker is Mg. **FM0098**



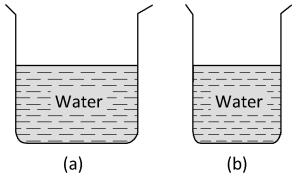
165. A massless conical flask filled with a liquid is kept on a table in a vacuum. The force exerted by the liquid on the base of the flask is W_1. The force exerted by the flask on the table is W_2.



(A) $W_1 = W_2$
(B) $W_1 > W_2$
(C) $W_1 < W_2$
(D) The force exerted by the liquid on the walls of the flask is $(W_1 - W_2)$. **FM0328**

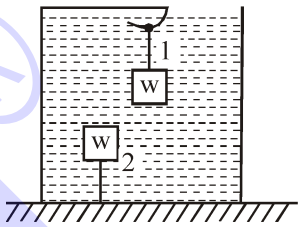
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

166. From the adjacent figure, the **incorrect** observation is



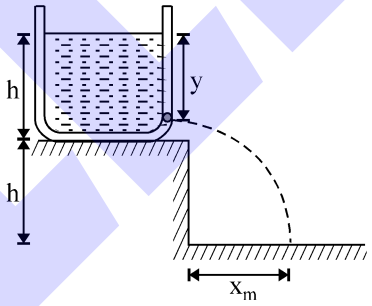
(A) The pressure on the bottom of tank (A) is greater than at the bottom of (B).
(B) The pressure on the bottom of the tank (A) is smaller than at the bottom of (B)
(C) The pressure depend on the shape of the container
(D) The pressure on the bottom of (A) and (B) is the same **FM0329**

167. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.



(A) iron block gets deflected towards right.
(B) wooden block gets deflected towards right.
(C) iron block gets deflected towards left.
(D) wooden block gets deflected towards left. **FM0099**

168. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then



(A) $x_m = 2h$ (B) $x_m = 1.5\ h$ (C) $y = h$ (D) $y = 0.75\ h$
FM0101

169. A small sphere of mass m is dropped from a height. After it has fallen 100m, it has attained its terminal velocity and continues to fall at that speed. Then the modulus of work done.
(A) by viscosity of air is lesser in first 100 m than in the second 100 m
(B) by buoyancy of air is in first 100 m is equal to that in the second 100 m
(C) by viscosity of air is greater in first 100 m than in the second 100 m
(D) by buoyancy of air is lesser in first 100 m than that in the second 100 m **FM0330**

170. A solid is completely immersed in a liquid. The force exerted by the liquid on the solid will
(A) increase if it is pushed deeper inside the liquid
(B) change if its orientation is changed
(C) decrease if it is taken partially out of the liquid
(D) be in the vertically upward direction **FM0331**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

171. When a capillary tube is dipped in a liqiud, the liquid rises to a height h in the tube. the free liquid surface inside the tube is hemispherical in shape. The tube is now pushed down so that the height of the tube outside the liquid is less than h . Then
(A) The liquid will come out of the tube like in a small fountain
(B) The liquid will ooze out of the tube slowly
(C) The liquid will fill the tube but not come out of its upper end
(D) The free liquid surface inside the tube will not be hemispherical **FM0332**

172. A solid sphere moves at a terminal velocity of 20 ms^{-1} in air at a place where g = 9.8 ms^{-2}. The sphere is taken in a gravity-free hall having air at the same pressure and pushed down at a speed of 20 ms^{-1} [Consider density of air to be negligible]
(A) Its initial acceleration will be 9.8 ms^{-2} downward.
(B) Its initial acceleration will be 9.8 ms^{-2} upward
(C) The magnitude of acceleration will decrease as the time passes.
(D) It will eventually stop. **FM0333**

173. Water is flowing in streamline motion through a tube with its axis horizontal. Consider two points A and B in the tube at the same horizontal level.
(A) The pressures at A and B are equal for any shape of the tube
(B) The pressures are never equal
(C) The pressures are equal if the tube has a uniform cross-section
(D) The pressuers may be equal even if the tube has a nonuniform corss-section. **FM0334**

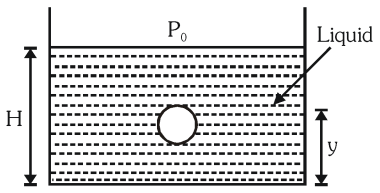
174. Water flows through two identical tubes A and B. A volume V_0 of water passes through the tube A and $2V_0$ through B in a given time. Which of the following may the correct ?
(A) Flow in both the tubes are steady
(B) Flow in both the tubes are turbulent.
(C) Flow is steady in A but turbulent in B
(D) Flow is steady in B but turbulent in A **FM0335**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 175 to 177

A small spherical monoatomic ideal gas bubble $\left(\gamma = \frac{5}{3}\right)$ is trapped inside a liquid of density ρ_ℓ (see figure). Assume that the bubble does not exchange any heat with the liquid. The bubble contains n moles of gas. The temperature of the gas when the bubble is at the bottom is T_0, the height of the liquid is H and the atmospheric pressure is P_0 (Neglect surface tension).

[IIT-JEE 2008]



node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

175. As the bubble moves upwards, besides the buoyancy force the following forces are acting on it.

(A) Only the force of gravity

(B) The force due to gravity and the force due to the pressure of the liquid

(C) The force due to gravity, the force due to the pressure of the liquid and the force due to viscosity of the liquid.

(D) The force due to gravity and the force due to viscosity of the liquid. **FM0150**

176. When the gas bubble is at a height y from the bottom, its temperature is :-

(A) $T_0\left(\frac{P_0+\rho_\ell gH}{P_0+\rho_\ell gy}\right)^{2/5}$ (B) $T_0\left(\frac{P_0+\rho_\ell g(H-y)}{P_0+\rho_\ell gH}\right)^{2/5}$

(C) $T_0\left(\frac{P_0+\rho_\ell gH}{P_0+\rho_\ell gy}\right)^{3/5}$ (D) $T_0\left(\frac{P_0+\rho_\ell g(H-y)}{P_0+\rho_\ell gH}\right)^{3/5}$ **FM0150**

177. The buoyancy force acting on the gas bubble is (Assume R is the universal gas constant)

(A) $p_\ell nRgT_0\frac{(P_0+\rho_\ell gH)^{2/5}}{(P_0+\rho_\ell gy)^{7/5}}$ (B) $\frac{\rho_\ell nRgT_0}{(P_0+\rho_\ell gH)^{2/5}[P_0+\rho_\ell g(H-y)]^{3/5}}$

(C) $p_\ell nRgT_0\frac{(P_0+\rho_\ell gH)^{3/5}}{(P_0+\rho_\ell gy)^{8/5}}$ (D) $\frac{\rho_\ell nRgT_0}{(P_0+\rho_\ell gH)^{3/5}[P_0+\rho_\ell g(H-y)]^{2/5}}$

FM0150

MATRIX MATCH TYPE QUESTION

Fluid Statics

178. Bucket A contains only water ; an identical bucket B contains water, but also contains a solid object in the water. Consider the following four situations. Which bucket weighs more ?

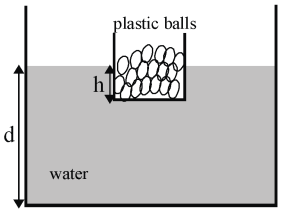
	Column-I		**Column-II**
(A)	The object floats in bucket B, and the buckets have the same water level	(P)	Bucket A
(B)	The object floats in bucket B, and the buckets have the same volume of water	(Q)	Bucket B
(C)	The object sinks completely in bucket B, and the buckets have the same water level	(R)	Both buckets have the same weight
(D)	The object sinks completely in bucket B, and the buckets have the same volume of water	(S)	The answer cannot be determined from the information given.

FM0102

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

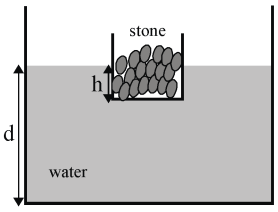
179. Column I shows different system as describe, with some parameter while column-II gives the change in the parameter.

Column-I **Column-II**

(A) 

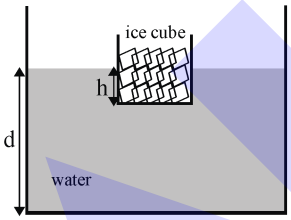
Plastic ball is thrown from the container in the water.
[d is the height of water level while h is the depth to which the container is submerged]

(P) h decreases

(B) 

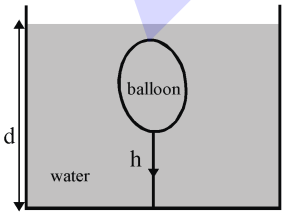
Stone is thrown from the container in the water.
[d is the height of water level while h is the depth to which the container is submerged]

(Q) h remain same

(C) 

Ice melts and remain in the container.
[d is the height of water level while h is the depth to which the container is submerged]

(R) h increases

(D) 

The water is heated slowly.
[d is the height of water level while h is the tension in the string. $\gamma_{air} > \gamma_{water}$]

(S) d decreases

(T) d remain same

FM0103'

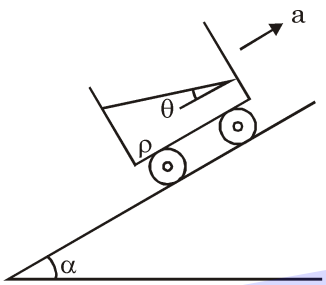
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

Fluid Statics

1. A fluid container is containing a liquid of densit8y ρ is accelerating upward with acceleration a along the inclined plane of inclination a as shown in the figure. Then the angle of inclination θ of free surface is

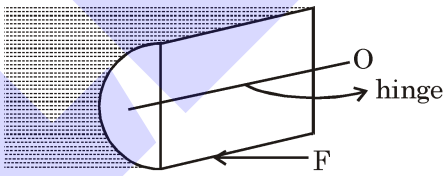


(A) $\tan^{-1}\left[\frac{a}{g\cos\alpha}\right]$ (B) $\tan^{-1}\left[\frac{a+g\sin\alpha}{g\cos\alpha}\right]$

(C) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1+\cos\alpha)}\right]$ (D) $\tan^{-1}\left[\frac{a-g\sin\alpha}{g(1-\cos\alpha)}\right]$

FM0336

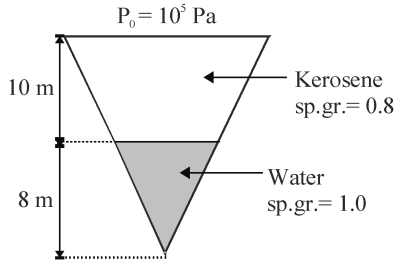
2. A light semi cylindrical gate of radius R is pivoted at its axis as shown in the figure holding liquid of density ρ. The force F required to prevent the rotation of the gate is equal to



(A) $2\pi R^3\rho g$ (B) $2\rho g R^3 l$ (C) $\frac{2R^2 l\rho g}{3}$ (D) none of these

FM0105

3. Shown in figure, a conical container of half-apex angle 37° filled with certain quantities of kerosene and water. The force exerted by the water on the kerosene is approximately, (Take atmospheric pressure = 10^5 Pa)



(A) 3×10^7 N (B) 4×10^7 N (C) 2×10^7 N (D) 5×10^7 N

FM0110

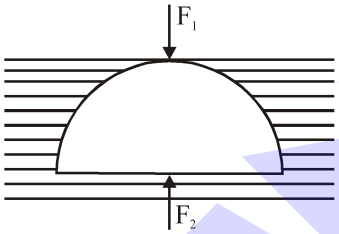
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

4. A heavy hollow cone of radius R and height h is placed on a horizontal table surface, with its flat base on the table. The whole volume inside the cone is filled with water of density ρ. The circular rim of the cone's base has a watertight seal with the table's surface and the top apex of the cone has a small hole. Neglecting atmospheric pressure find the total upward force exerted by water on the cone is :

(A) $(2/3)\pi R^2h\rho g$ (B) $(1/3)\pi R^2h\rho g$ (C) $\pi R^2h\rho g$ (D) None

FM0106

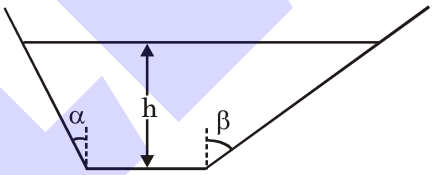
5. A solid hemisphere is just pressed below the liquid, the value of $\frac{F_1}{F_2}$ is (where F_1 and F_2 are the hydrostatic forces acting on the curved and flat surfaces of the hemisphere) (Neglect atmospheric pressure).



(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) none of these

FM0057

6. A liquid is filled upto height h in a vessel, as shown. find correct option (s);



(A) If $\alpha = \beta$, horizontal component of forces on left and right side of inclined faces will be equal and opposite.

(B) If $\alpha \neq \beta$, horizontal component of forces on left and right side of inclined faces will be equal and opposite.

(C) If A is the area of the base of the vessel, then force exerted by liquid on the vessel is greater than $(P_{atm} + \rho gh)A$.

(D) As above, the force exerted by liquid on walls is equal to $(P_{atm} + \rho gh)A$. **FM0337**

7. A closed cubical box is completely filled with water and is acceleration horizontally towards right with an acceleration a. The resultant normal force by the water on the top of the box

(A) passes through the cnetre of the top

(B) passes throught a point to the right of the centre

(C) passes through a point to the left of the centre

(D) becomes zero. **FM0338**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

Buoyancy

8. Two cubes of size 1.0 m sides, one of relative density 0.60 and another of relative density = 1.15 are connected by weightless wire and placed in a large tank of water. Under equilibrium the lighter cube will project above the water surface to a height of

(A) 50 cm (B) 25 cm (C) 10 cm (D) zero

FM0107

9. Mercury is poured into a uniform vertical U-tube and water is poured in above it. The level of water is the same in both arms. A piece of wood is dropped into one arm and some water equal in weight to the piece of wood is added to the other. Then, consider the statements :

(I) The level of mercury will be same in both the arms.

(II) The level of water will be same in both the arms.

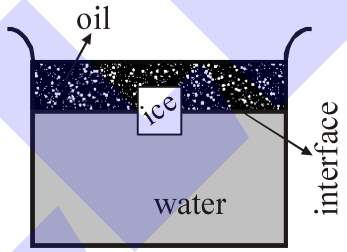
(A) Both I and II are true (B) I is true but II is false

(C) I is false but II is true (D) Both I and II are false. **FM0109**

10. A cube (density 0.5 gm/cc) of side 10 cm is floating in water kept in a cylindrical beaker of base area 1500 cm^2. When a mass m is kept on wooden block the level of water rises in the beaker by 2mm. Find the mass m.

(A) 200 g (B) 250 g (C) 300 g (D) 350 g **FM0339**

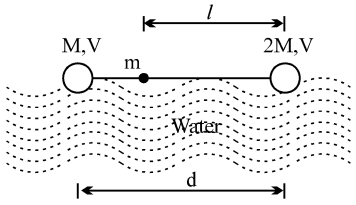
11. An ice cube is floating in water above which a layer of a lighter oil is poured. As the ice melts completely, the level of interface and the upper most level of oil will respectively :-



(A) rise and fall (B) fall and rise

(C) rise and not change (D) not change and fall **FM0111**

12. A dumbbell is placed in water of density ρ. It is observed that by attaching a mass m to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged as shown in the figure. The volume of the mass m is negligible. The value of length *l* is :-

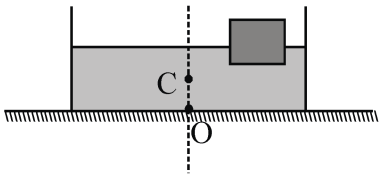


(A) $\frac{d(V\rho-3M)}{2(V\rho-2M)}$ (B) $\frac{d(V\rho-2M)}{2(V\rho-3M)}$ (C) $\frac{d(V\rho+2M)}{2(V\rho-3M)}$ (D) $\frac{d(V\rho-2M)}{2(V\rho+3M)}$

FM0123

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

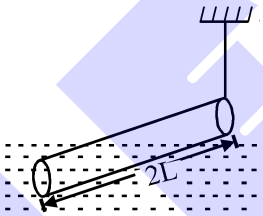
13. There is water in container with center of mass at C. Now a small wooden piece is place towards right as shown in the figure. After putting the wooden piece.



(A) Pressure at base remains same and centre of mass of water and wooden piece will be right of line OC.
(B) Pressure at base remains same and centre of mass of water and wooden piece will be on line OC.
(C) Pressure at base changes and centre of mass of water and wooden piece will be right of line OC.
(D) Pressure at base changes and centre of mass of water and wooden piece will be on line OC.

FM0120

14. A slender homogeneous rod of length $2L$ floats partly immersed in water, being supported by a string fastened to one of its ends, as shown. The specific gravity of the rod is 0.75. The length of rod that extends out of water is :



(A) L (B) $\frac{1}{2}$ L (C) $\frac{1}{4}$ L (D) 3 L

FM0108

15. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_C is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is **[IIT-JEE-2012]**

(A) more than half-filled if ρ_C is less than 0.5
(B) more than half-filled if ρ_C is more than 1.0
(C) half-filled if ρ_C is more than 0.5
(D) less than half-filled if ρ_c is less than 0.5

FM0155

16. An object of density d_o kept deep inside water of density d_w and released. During the time it moves a vertical distance h with in the water :-

(A) The gravitational potential energy of the water in the vessel increases if $d_0 < d_w$
(B) The gravitational potential energy of the water in the vessel decreases if $d_0 < d_w$
(C) The gravitational potential energy of the object increases if $d_0 > d_w$
(D) The gravitational potential energy of the object decreases if $d_0 < d_w$

FM0066

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

17. A piece of metal weighs 46 g in air and 30 g in a liquid of density 1.24×10^3 kgm^{-3} kept at 27°C. When the temperature of the liquid is raised to 42°C, the metal piece weighs 30.5 g. The density of the liquid at 42°C is 1.20×10^3 kgm^3. Calculate the coefficient of linear expansion of the metal.

(A) 1.8×10^{-5} °C^{-1} (B) 2.3×10^{-5} °C^{-1}

(C) 2.8×10^{-5} °C^{-1} (D) 3.0×10^{-5} °C^{-1} **FM0340**

18. A metal cube is placed in empty vessel. When water filled in the vessel so that the cube is completely mersed in the water, the force on the bottom of the vessel in contact with the cube

(A) will increase (B) will decrease

(C) will remain the same (D) will become zero **FM0341**

Surface Tension

19. If a soap bubble of radius 3 cm coalesce with another soap bubble of radius 4 cm under isothermal conditions, the radius of the resultant bubble formed is (in cm). (Both bubbles are in vacuum).

(A) 7 (B) 1 (C) 5 (D) 12 **FM0342**

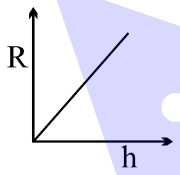
20. Calculate the force required to separate the glass plates of area 10^{-2} m^2 with a film of water 0.05 mm thickness between them (surface tension of water = 70×10^{-3} N/m) (contact angle = 0°).

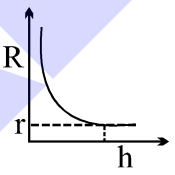
(A) 28 N (B) 112 N (C) 5.6 N (D) 11.2 N **FM0343**

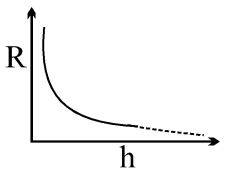
21. Two large glass plates are placed vertically and parallel to each other inside a tank of water with separation between the plates equal to 1 mm. Find the rise of water in the space between the plates. Surface tension of water = 0.075 N/m. (Contact angle = 0°).

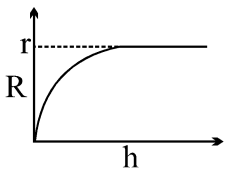
(A) 1.5 cm (B) 1.8 cm (C) 2.5 cm (D) 3.0 cm **FM0344**

22. A long capillary tube of radius 'r' is initially just vertically completely immerged inside a liquid of angle of contact 0^0. If the tube is slowly raised then relation between radius of curvature of meniscus inside the capillary tube and displacement (h) of tube can be represented by

(A) 

(B) 

(C) 

(D) 

FM0112

Fluid Dynamics

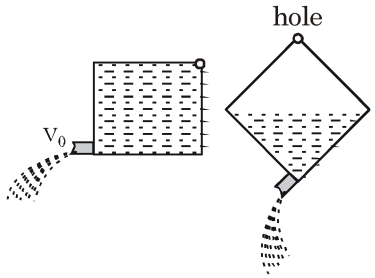
23. Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate 3 ms^{-1}. The enclosure has 100 holes each of diameter 0.05 cm. The velocity of water coming out of the holes is (in ms^{-1}):

(A) 0.48 (B) 96 (C) 24 (D) 48

FM0124

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

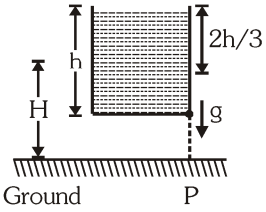
24. A cubical box of wine has a small spout located in one of the bottom corners. When the box is full and placed on a level surface, opening the spout results in a flow of wine with a initial speed of v_0 (see figure). When the box is half empty, someone tilts it at 45° so that the spout is at the lowest point (see figure). When the spout is opened the wine will flow out with a speed of



(A) v_0 (B) $v_0/2$ (C) $v_0/\sqrt{2}$ (D) $v_0/\sqrt[4]{2}$

FM0117

25. An open vessel full of water is falling freely under gravity. There is a small hole in one face of the vessel, as shown in the figure. The water which comes out from the hole at the instant when hole is at height H above the ground, strikes the ground at a distance of x from P. Which of the following is correct for the situation described ?



(A) The value of x is $2\sqrt{\frac{2hH}{3}}$

(B) The value of x is $\sqrt{\frac{4hH}{3}}$

(C) The value of x can't be computed from information provided.

(D) The question is irrevalent as no water comes out from the hole. **FM0125**

26. Water coming out of a horizontal tube at a speed v strikes normally a vertically wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to 2v.

(A) the thrust exerted by the water on the wall will be doubled

(B) the thrust exerted by the water on the wall will be unchanged

(C) the energy lost per second by water striking the wall will also be four times

(D) the energy lost per second by water striking the wall be increased eight times. **FM0115**

27. A vertical tank, open at the top, is filled with a liquid and rests on a smooth horizontal surface. A small hole is opened at the centre of one side of the tank. The area of cross-section of the tank is N times the area of the hole, where N is a large number. Neglect mass of the tank itself. The initial acceleration of the tank is :-

(A) $\frac{g}{2N}$ (B) $\frac{g}{\sqrt{2N}}$ (C) $\frac{g}{N}$ (D) $\frac{g}{2\sqrt{N}}$

FM0128

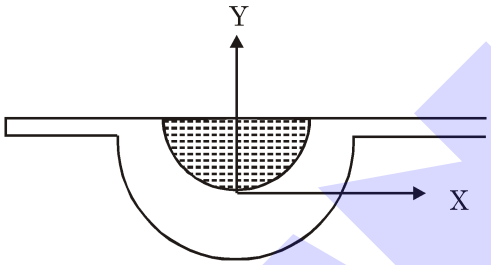
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

28. A tank full of water has a small hole at the bottom. If one-fourth of the tank is emptied in t_1 seconds and the remaining three-fourths of the tank is emptied in t_2 seconds. Then the ratio $\frac{t_1}{t_2}$ is

(A) $\sqrt{3}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{2}{\sqrt{3}}-1$

FM0345

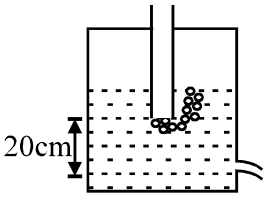
29. A small hole is made at the bottom of a symmetrical jar as shown in figure. A liquid is filled in to the jar up to a certain height. The rate of descent of liquid is Independent of level of the liquid in the jar. Then the surface of jar is a surface of revolution of curve



(A) $y = kx^4$ (B) $y = kx^2$ (C) $y = kx^3$ (D) $y = kx^5$

FM0346

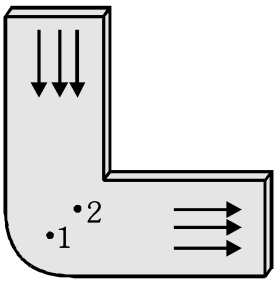
30. A tube is attached as shown in closed vessel containing water. The velocity of water coming out from a small hole is :



(A) $\sqrt{2}$ m/s
(B) 2 m/s
(C) depends on pressure of air inside vessel
(D) None of these

FM0075

31. Ideal fluid flows along a flat tube of constant cross-section, located in a horizontal plane and bent as shown in Fig. (top view). The flow is steady. Select correct option regarding the pressures and velocities of the fluid at points 1 and 2.



(A) $p_1 > p_2$, $v_1 > v_2$
(B) $p_1 < p_2$, $v_1 < v_2$
(C) $p_1 > p_2$, $v_1 < v_2$
(D) $p_1 < p_2$, $v_1 > v_2$

FM0347

32. Two balls of same size but different masses m_1 and m_2 ($m_2 > m_1$) are attached to the two ends of a thin light thread and dropped from a certain height. It is known that the viscous drag of air depends on the size and velocities of the balls. Other than the gravitational pull from the earth and the viscous drag, the buoyant force from air also act on the balls. The buoyant force on a ball equals to the weight of air displaced by the ball. After sufficiently long time from the instant the balls were dropped both of them acquire uniform velocity known as terminal velocity. When the balls have acquired terminal velocity, the tension in the thread is

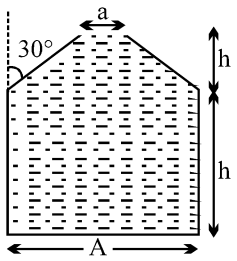
(A) Zero (B) $(m_2 - m_1)g$ (C) $0.5(m_2 + m_1)g$ (D) $0.5(m_2 - m_1)g$

FM0348

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

MULTIPLE CORRECT TYPE QUESTIONS

33. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h. The upper part is a conical vessel of height h with base area 'A' and top area 'a' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height 2h. Neglecting atmospheric pressure.



(A) The force F exerted by the liquid on the base of the vessel is $2h\rho g\,\frac{(A+a)}{2}$.

(B) the pressure P at the base of the vessel is $2h\rho g\,\frac{A}{a}$.

(C) the weight of the liquid W is greater than the force exerted by the liquid on the base

(D) the walls of the vessel exert a downward force (F–W) on the liquid. **FM0104**

34. The weight of an empty balloon on a spring balance is W_1. The weight becomes W2 when the ballon is filled with air. Let the weight of the air itself be w. Neglect the thickness of the ballon when it is filled with air. Also neglect the difference in the densities of the air inside and outside the ballon.

(A) $W_2 = W_1$ (B) $W_2 = W_1 + \omega$

(C) $W_2 < W_1 + \omega$ (D) $W_2 > W_1$ **FM0349**

35. A cubical block is floating in a liquid with one third of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2 :-

(A) the fraction of volume immersed in the liquid will change.

(B) the buoyancy force on the block will change.

(C) the buoyancy force will increase by 50 percent.

(D) the pressure in the liquid will increased. **FM0132**

COMPREHENSION TYPE QUESTIONS

Paragraph for Questions no. 36 to 38

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R. When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper. **[IIT-JEE-2010]**

36. If the radius of the opening of the dropper is r, the vertical force due to the surface tension on the drop of radius R (assuming r <<R) is

(A) 2πrT (B) 2πRT (C) $\frac{2\pi r^2 T}{R}$ (D) $\frac{2\pi R^2 T}{r}$ **FM0153**

37. If r = 5×10^{-4} m, ρ = 10^3 kgm^{-3}, g = 10 ms^{-2}, T = 0.11 Nm^{-1}, the radius of the drop when it detaches from the dropper is approximately

(A) 1.4×10^{-3} m (B) 3.3×10^{-3} m (C) 2.0×10^{-3} m (D) 4.1×10^{-3} m

FM0153

38. After the drop detaches, its surface energy is

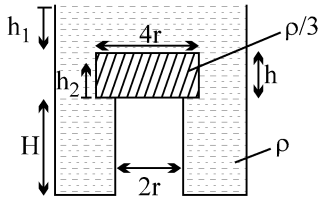
(A) 1.4×10^{-6} J (B) 2.7×10^{-6} J (C) 5.4×10^{-6} J (D) 8.1×10^{-6} J

FM0153

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

Paragraph for Question no. 39 to 41

A wooden cylinder of diameter 4r, height h and density ρ/3 is kept on a hole of diameter 2r of a tank, filled with water of density ρ as shown in the figure. The height of the base of cylinder from the base of tank is H. **[IIT-JEE 2006]**



39. If level of liquid starts decreasing slowly when the level of liquid is at a height h_1 above the cylinder, the block just starts moving up. Then, value of h_1 is :-

(A) $\frac{2h}{3}$ (B) $\frac{5h}{4}$ (C) $\frac{5h}{3}$ (D) $\frac{5h}{2}$ **FM0133**

40. Let the cylinder is prevented from moving up, by applying a force and water level is further decreased. Then, height of water level (h_2 in figure) for which the cylinder remains in original position without application of force is

(A) $\frac{h}{3}$ (B) $\frac{4h}{9}$ (C) $\frac{2h}{3}$ (D) h **FM0133**

41. If height h_2 of water level is further decreased, then

(A) cylinder will not move up and remains at its original position.

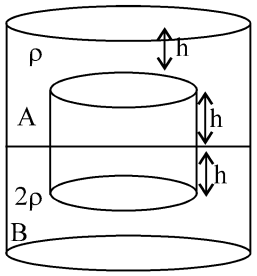
(B) for $h_2 = h/3$, cylinder again starts moving up

(C) for $h_2 = h/4$, cylinder again starts moving up

(D) for $h_2 = h/5$ cylinder again starts moving up **FM0133**

MATRIX MATCH TYPE QUESTION

42. Shown below is a cylinder of radius R floating in vessel containing liquids A and B. Neglecting atmospheric pressure match the quantities mentioned in column-I with corresponding expression in column-II.

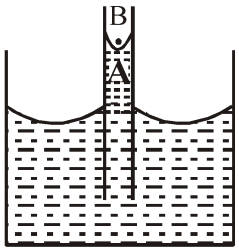
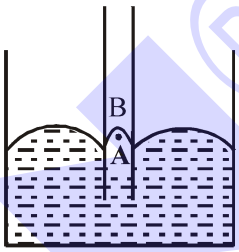
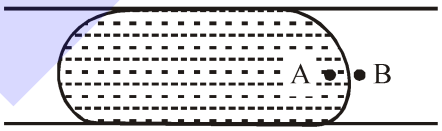
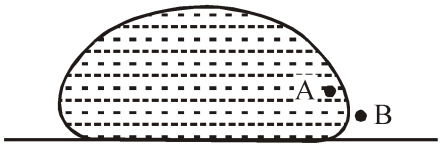
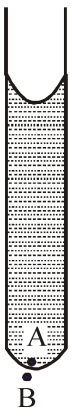


	Column-I		Column-II
(A)	Net force exerted by liquid A of density ρ on the cylinder.	(P)	$9\ \rho g R h^2$
(B)	Net force exerted by liquid B of density 2ρ on the cylinder.	(Q)	$\pi\rho g R^2 h$
(C)	Net force exerted by liquids A and B on the left half of the curved part of cylinder.	(R)	$4\pi\rho g R^2 h$
(D)	Net force exerted by liquid A and B on the cylinder.	(S)	$3\pi\rho g R^2 h$

FM0134

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

43. Capillary rise and shape of droplets on a plate due to surface tension are shown in column-II. Point A and B are just inside and outside the surface. Match the following

Column-I	**Column-II**
(A) liquid is wetting the solid	(P) 
(B) liquid is not wetting the solid	(Q) 
(C) Pressure at A > pressure at B	(R) A liquid drop is pressed between two parallel glass plates. 
(D) Pressure at B > Pressure at A	(S) 
	(T)  both end open cylindrical pipe

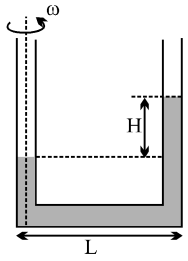
FM0166

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

EXERCISE (S)

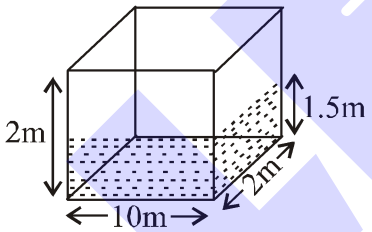
Fluid Statics

1. A U tube is rotated about one of it's limbs with an angular velocity ω. Find the difference in height H of the liquid (density ρ) level, where diameter of the tube d << L. **[IIT-JEE 2005]**



FM0007

2. An open tank 10 m long and 2 m deep is filled upto height 1.5 m of oil of specific gravity 0.80. The tank is accelerated uniformly from rest to a speed of 10 m/sec towards right. What is the shortest time in which this speed may be attained without spilling any oil. [g = 10m/s^2]:



FM0350

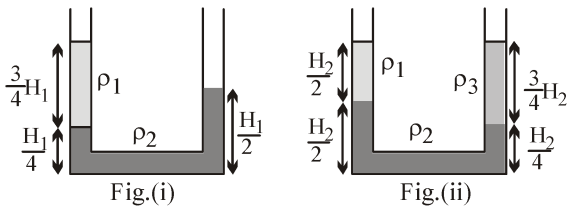
3. The volume of an air bubble is doubled as it rises from the bottom of a lake to its surface. If the atmospheric pressure is H m of mercury & the density of mercury is n times that of lake water. Find the depth of the lake. Neglect surface tension.

FM0011

4. A U-tube filled with a liquid of volumetric coefficient of 10^{-5}/°C lies in a vertical plane. The height of liquid column in the left vertical limb is 100 cm. The liquid in the left vertical limb is maintained at a temperature = 0°C while the liquid in the right limb is maintained at a temperature = 100°C. Find the difference in levels in the two limbs.

FM0038

5. There are three immiscible liquids of densities ρ_1, ρ_2 and ρ_3. If the first two liquids are taken in a U–tube then they stand like fig(i). When all the three liquids are taken in U–tube, they stand like fig(ii). Find the ratio of the densities of the three liquids.



FM0039

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

6. Dams at two different locations are needed to form a lake. When the lake is filled, the water level will be at top of both dams. The second dam is twice as high and twice as wide as the first dam. The force of the water on the second dam is how much greater than the force on the first? (Ignore atmospheric pressure since it is pushing on both sides of the dams.)

FM0005

7. Water is filled in a rectangular tank of size 3 m × 2 m × 1 m. 1m is the height.

(a) Find the total force exerted by the water on the bottom surface of the tank.

(b) Consider a vertical side of area 2 m × 1 m. Take a horizontal strip of width δx metre in this side, situated at a depth of x metre from the surface of water. Find the force by the water on this strip.

(c) Find the torque of the force calculated in part (b) about the bottom edge of this side.

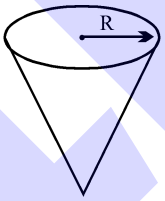
(d) Find the total force by the water on this side.

[e] Find the total torque by the water on the side about the bottom edge. Neglect the atmospheric pressure and take $g = 10\ m/s^2$. **FM0351**

8. An inverted hollow cone, of height H and radius at the top = R, is filled completely with a liquid of density d. Neglect atmosphere pressure.

(a) Find horizontal component of the force which the liquid exerts on half curved side wall of the cone.

(b) Find the vertical downward component of the thrush force due to water pressure which the liquid exerts on half side curved wall of the cone.

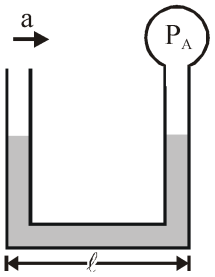


FM0037

9. A spherical tank of 1.2 m radius is half filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of $10\ m/s^2$. Calculate the inclination of the oil surface to horizontal and maximum pressure on the tank.

FM0001

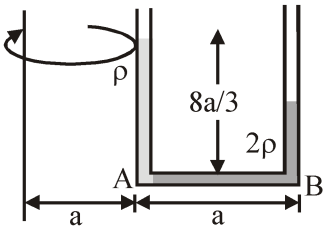
10. A liquid of density ρ is filled in a U-tube, whose one end is open & at the other end a bulb is fitted whose pressure is P_A. Now this tube is moved horizontally with acceleration 'a' as shown in the figure. During motion it is found that liquid in both column is at same level at equilibrium. If atmospheric pressure is P_0, then find the value of P_A



FM0003

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

11. The interface of two liquids of densities ρ and 2ρ respectively lies at the point A in a U tube at rest. The height of liquid column above A is 8a/3 where AB = a. The cross sectional area of the tube is S. With what angular velocity the tube must be whirled about a vertical axis at a distance 'a' such that the interface of the liquids shifts towards B by 2a/3.

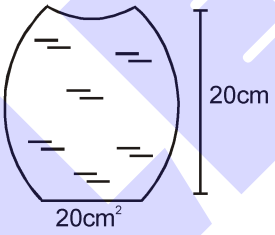


FM0040

12. A cylindrical vessel with water is rotated about its vertical axis with a constant angular velocity ω. Find:
(a) the shape of the free surface of the water;
(b) the water pressure distribution over the bottom of the vessel along its radius provided the pressure at the central point is equal to p_0.

FM0352

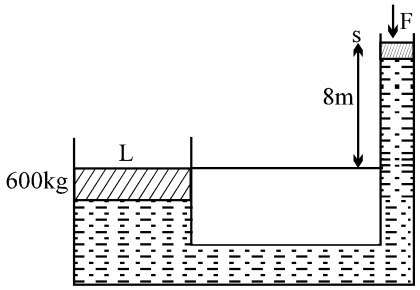
13. A glass full of water has a bottom of area 20 cm^2, top of area 20 cm^2, height 20 cm and volume half a litre.



(a) Find the force exerted by the water on the bottom.
(b) Considering the equilibrium of the water, find the resultant force exerted by the sides of the glass on the water. Atmospheric pressure = 1.0×10^5 N/m^2. Density of water = 1000 kg/m^3 and g = 10 m/s^2. Take all numbers to be exact.

FM0353

14. For the system shown in the figure, the cylinder on the left at L has a mass of 600 kg and a cross sectional area of 800 cm^2. The piston on the right, at S, has cross sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil.($\rho = 0.75$ gm/cm^3) Find the force F required to hold the system in equilibrium.



FM0004

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

Buoyancy

15. In air an object weighs 15N, when immersed completely in water the same object weighs 12N. When immersed in another liquid completely, it weighs 13N. Find
(a) the specific gravity of the object and
(b) the specific gravity of the other liquid. **FM0009**

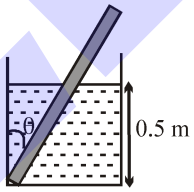
16. A hollow cone floats with its axis vertical upto one-third of its height in a liquid of relative density 0.8 and with its vertex submerged. When another liquid of relative density ρ is filled in it upto one-third of its height, the cone floats upto half its vertical height. The height of the cone is 0.10 m and the radius of the circular base is 0.05 m. Find the specific gravity ρ. **FM0027**

17. A cubical metal block of edge 12 cm floats in mercury with one fifth of the height inside the mercury. Water is poured till the surface of the block is just immersed in it. Find the height of the water column to be poured. Specific gravity of mercury = 13.6. **FM0354**

18. A ferry boat has internal volume 1 m^3 and weight 50 kg.
(a) Neglecting the thickness of the wood, find the fraction of the volume of the boat immersed in water.
(b) If a leak develops in the bottom and water starts coming in, what fraction of the boat's volume will be filled with water before water starts coming in from the sides ? **FM0355**

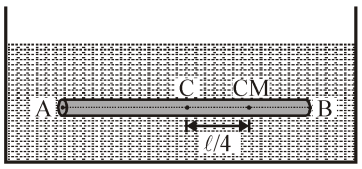
19. An expansible balloon filled with air floats on the surface of a lake with 2/3 of its volume submerged. How deep must it be sunk in the water so that it is just inequilibrium neither sinking further nor rising? It is assumed that the temperature of the water is constant & that the height of the water barometer is 9 meters. **FM0041**

20. A wooden plank of length 1 m and uniform cross-section is hinged at one end to the bottom of a tank as shown. The tank is filled with water upto a height of 0.5 m. The specific gravity of the plank is 0.5. Find the angle θ made by the plank in equilibrium position



FM0010

21. A non uniform cylinderical rod of mass 5kg, length 2m and radius 0.1m is having its center of mass at a distance $\frac{1}{2}$m from the centre and lying on the axis of the rod. The rod is kept in a liquid of uniform density 1400 kg/m^3. The moment of inertia of the rod about the centre of mass is 4 kg-m^2. Find the angular acceleration (in SI unit) of point A of the rod just after the rod is released from the position shown in figure.



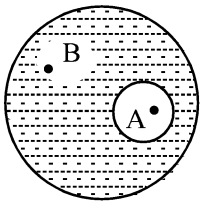
FM0356

22. A block of ice with an area of cross section A and a height h floats in water of density ρ_0. What work should be performed to submerge the ice block completely into water if density of ice is ρ_1? **FM0357**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

Surface Tension

23. There is an air bubble of radius R inside a drop of water of radius 3R. Find the ratio of gauge pressure at point A to the gauge pressure at point B.

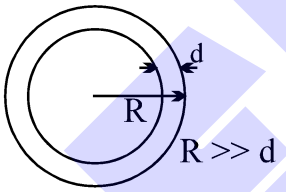


FM0012

24. There is a soap bubble of radius 2.4×10^{-4}m in air cylinder which is originally at a pressure of 10^5 N/m^2 .The air in the cylinder is now compressed isothermally untill the radius of the bubble is halved.(The surface tension of the soap film is 0.08Nm^{-1}).The pressure of air in the cylinder is found to be 8.08×10^n N/m^2. What is the value of n?

FM0358

25. A soap bubble has radius R and thickness d (<< R) as shown. It collapses into a spherical drop. Find the ratio of excess pressure in the drop to the excess pressure inside the bubble.

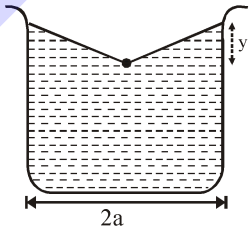


FM0042

26. The diameter of a gas bubble formed at the bottom of a pond is d = 4.0 μm. When the bubble rises to the surface its diameter increases n = 1.1 times. Find how deep is the pond at that spot. The atmospheric pressure is standard, the gas expansion is assumed to be isothermal. Take S = 0.08 N/m

FM0359

27. A container of width 2a is filled with a liquid. A thin wire of weight per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance y (y << a). Determine the surface tension of the liquid. **[2004, 2M]**



FM0044

28. The lower end of a capillary tube of radius 1 mm is dipped vertically into mercury.
(a) Find the depression of mercury column in the capillary.
(b) If the length dipped inside is half the answer of part (a), find the angle made by the mercury surface at the end of the capillary with the vertical. Surface tension of mercury =0.465 N/m and the contact angle of mercury with glass = 135^0. **FM0360**

29. A drop of water volume 0.05 cm^3 is pressed between two glass-plates, as a consequence of which, it spreads between the plates. The area of contact with each plate is 40 cm^2. If the surface tension of water is 70 dyne/cm, the minimum normal force required to seperate out the two glass plates in newton is approximately (assuming angle of contact is zero) : **FM0361**

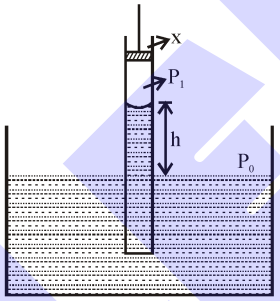
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

30. A capillary tube of radius 1 mm is kept vertical with the lower end in water. S = 0.075 N/m, θ = 0°
(a) find the height of water raised in the capillary.
(b) If the length of the capillary tube is half the answer of part (a), find the angle θ made by the water surface in the capillary with the wall. **FM0362**

31. The lower end of a capillary tube is immersed in mercury. The level of mercury in the tube is found to be 2 cm below the outer level. If the same tube is immersed in water, upto what height will the water rise in the capillary ? (Density of mercury = 13.6 gm/cm^3, density of water = 1 gm/cm^3, surface tension of mercury = 484.5 dyne/cm, surface tension of water = 72.7 dyne/cm, angle of contact between mercury & glass = 135°)

FM0363

32. A capillary of radius r = 0.2 mm is dipped vertically in a liquid of density, (ρ = 1gm / cc). A small piston (x) is inserted inside the capillary which maintains the constant pressure of air ($P_1 = 2 \times 10^4$ dynes/cm^2) above the hemispherical meniscus. If ambient pressure is $P_0 = 10^5$ dynes/cm^2 and surface tension of liquid is T = 72 dynes/cm, then find the height h of liquid in cm. (weight of curved part of liquid is ignored)

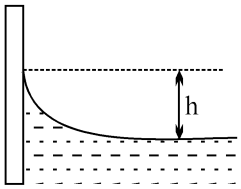


FM0043

33. When a vertical capillary of length ℓ with the sealed upper end was brought in contact with the surface of a liquid, the level of this liquid rose to the height h. The liquid density is ρ, the inside diameter of the capillary is d, the contact angle is θ, the atmospheric pressure is p_0, Find the surface tension of the liquid. **FM0364**

34. Capillary tube with very thin walls is suspended vertically in a cup containing water. Before the capillary is suspended, the cup was balanced against weights on a beam balance. When capillary is dipped to a small distance in the water, water rises in the capillary. Water completely wets the capillary walls. To restore the balance, we have to decrease the load on the other scales by 0.14 g. If the surface tension of water is 0.07 N/m, determine the radius r (in mm) of the capillary. **FM0365**

35. Water in a glass aquarium forms a meniscus, as illustrated in the figure. Show that the difference in height h between the centre and the edge of the meniscus is given by $h = \sqrt{\frac{2\gamma}{\rho g}} \approx 4$ mm. The surface tension of water is $\gamma = 0.073$ Nm^{-1}. Wetting is complete.

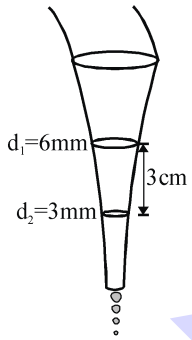


FM0366

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

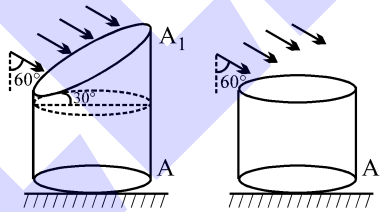
Fluid Dynamics

36. The tap in the garden was closed in appropriately resulting in the water flowing freely out of it which forms a downward narrowing beam. The beam of water has a circular cross-section, the diameter of the circle is 6 mm at one point and 3 cm below it is only 3 mm as shown in figure. If the rate of water wasted is $(x \times \pi)$ mL/minute then find the value of x. (Neglect the effect of viscosity and surface tension of the flowing water.)



FM0021

37. A vertical cylindrical container of base area A and upper cross-section area A_1 making an angle 30° with the horizontal is placed in an open rainy field as shown near another cylindrical container having same base area A.Find the ratio of rates of collection of water in the two containers.

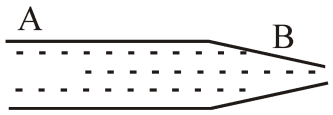


FM0045

38. Calculate the rate of flow of glycerine of density 1.25 x 10^3 kg/m^3 through the conical section of a pipe if the radii of its ends are 0.1m & 0.04m and the pressure drop across its length is 10N/m^2.

FM0017

39. Water flows through a horizontal tube of variable cross-section (figure). The area of cross-section at A and B are 4 mm^2 and 2 mm^2 respectively. If 1 cc of water enters per second through A, find



(a) the speed of water at A,

(b) the speed of water at B and

(c) the pressure difference $P_A - P_B$

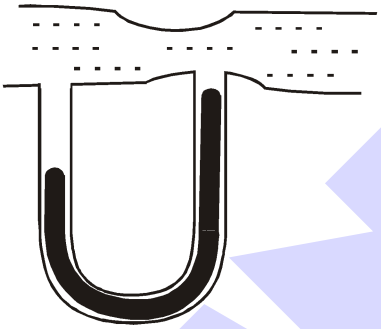
FM0367

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

40. Suppose the tube in the previous problem is kept vertical with A upward but the other condition remain the same. The separation between the cross-section at A and B is $\frac{15}{16}$ cm. Repeat parts (a), (b) and (c) of the previous problem. Take g = 10 m/s^2. **FM0368**

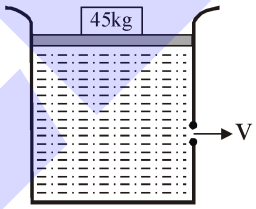
41. Suppose the tube in the previous problem is kept vertical with B upward. Water enters through B at the rate of 1 cm^3/s. Repeat parts (a), (b) and (c). Note that the speed decreases as the water falls down. **FM0369**

42. Water flows through the tube shown in figure. The areas of cross-section of the wide and the narrow portions of the tube are 5 cm^2 and 2 cm^2 respectively. The rate of flow of water through the tube is 500 cm^3/s. Find the difference of mercury levels in the U-tube. **FM0370**

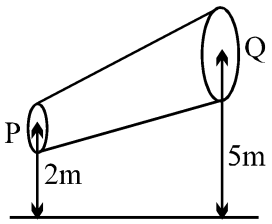


43. A jet of water having velocity = 10 m/s and stream cross-section = 2 cm^2 hits a flat plate perpendicularly, with the water splashing out parallel to plate. Find the force that the plate experiences. **FM0015**

44. A large cylindrical tank of cross-sectional area 1m^2 is filled with water. It has a small hole at a height of 1m from the bottom. A movable piston of mass 5 kg is fitted on the top of the tank such that it can slide in the tank freely. A load of 45 kg is applied on the top of water by piston, as shown in figure. Find the value of v when piston is 7m above the bottom (g = 10 m/s^2) **FM0019**



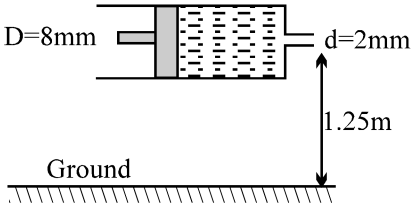
45. A non-viscous liquid of constant density 1000 kg/m^3 flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively 4×10^{-3}m^2 and 8×10^{-3} m^3. The velocity of the liquid at point P is 1 m/s. Find the work done per unit volume by the pressure and the gravity forces as the fluid flows from point P to Q.



FM0034

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

46. Consider a horizontally oriented syringe containing water located at a height of 1.25 m above the ground. The diameter of the plunger is 8 mm and the diameter of the nozzle is 2 mm. The plunger is pushed with a constant speed of 0.25 m/s. Find the horizontal range of water stream on the ground. Take g = 10 m/s^2. **[IIT-JEE 2004]**



FM0032

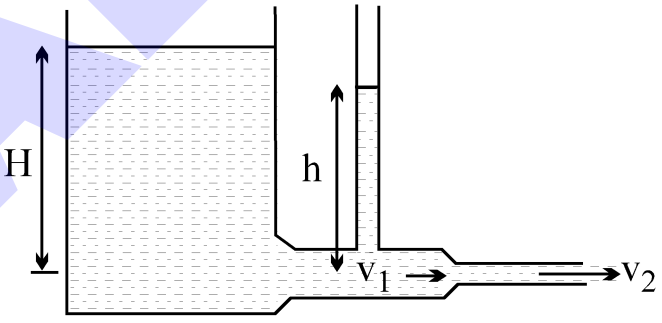
47. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H. Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice. [Take atmospheric pressure = 1.0×10^5 N/m^2, density of water = 1000 kg/m^3 and g = 10 m/s^2. Neglect any effect of surface tension.] **[IIT-JEE-2009]**

FM0151

48. A wide vessel with small hole in the bottom is filled with water and kerosene. Neglecting viscosity, find the velocity of water flow v, if the thickness of water layer is h_1 and that of kerosene layer is h_2 is (density of water ρ_1 gm/cc and that of kerosene is ρ_2 gm/cc.

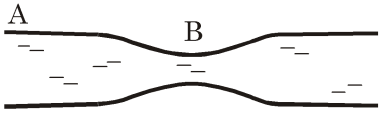
FM0046

49. Water fills a reservoir, open to the atmosphere, to height H = 5m above the centerline of a horizontal exit pipe at the bottom of the reservoir. The first section of the pipe has radius r_1 = 2 cm, the unknown velocity there is v_1 and this section of the pipe has a manometer in which the water rises to an unknown height h above the centerline of the pipe. The water leaves the second section of radius r_2 = 1 cm, with speed v_2. Assume the flow is incompressible, frictionless, irrotational, and steady. Also, assume the reservoir is so large compared to the pipe, that the water level in the reservoir is almost constant.
(a) What is the speed, v_2, of the water leaving the exit pipe?
(b) What is the speed, v_1, in the first section of the pipe?
(c) What is the height, h, of the water in the manometer?
(d) What is the volume flow rate of the water flowing through the exit pipe?



FM0048

50. Water flows in a horizontal tube as shown in figure. The pressure of water changes by 600 N/m^2

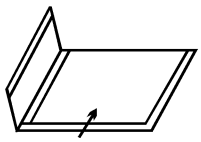


between A and B where the areas of cross-section are 30 cm^2 and 15 cm^2 respectively. Find the rate of flow of water through the tube. **FM0371**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

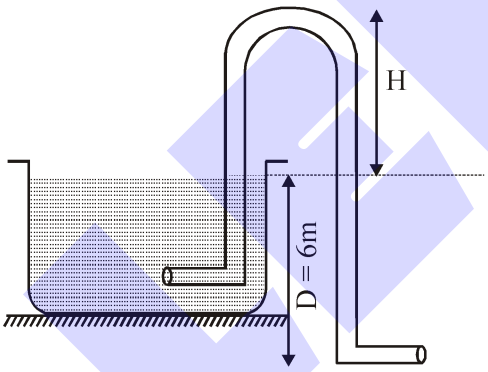
51. A glass plate of area A and mass m is hinged along one of its sides. At what speed the air should be blown parallel to its upper surface to hold the plate horizontal? The density of air is ρ. Give the answer in term of

$$\sqrt{\frac{mg}{18\rho A}}$$



FM0049

52. Figure shows a siphon, the vessel area is very large as compared to cross section of tube. Tube has a uniform cross section, its lower end is 6m below the surface of water. What is maximum height H (in m) of the upper end for siphon to work ? Take density of water = 10^3 kg/m^3, atmospheric pressure = 10^5 N/m^2.



FM0050

Viscosity

53. A tube of length ℓ and radius R carries a steady flow of fluid whose density is ρ and viscosity η. The fluid flow velocity depends on the distance r from the axis of the tube as $v = v_0 (1 - r^2/R^2)$. Find:
(a) the volume of the fluid flowing across the section of the tube per unit time;
(b) the kinetic energy of the fluid within the tube's volume;
(c) the friction force exerted on the tube by the fluid;
(d) the pressure difference at the ends of the tube. **FM0372**

54. Two large vertical parallel plates separated by a gap of d = 1mm have a highly viscous liquid of density 800 kg/m^3 and viscosity coefficient η = 5 poise flowing steadily under gravity in between the gap.Find the velocity gradient of flow near plates surface in sec^{-1} also find force of viscosity per unit area on each of plates. (1 poise = 0.1 NS/m^2).

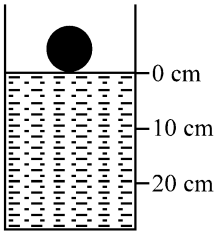
FM0036

55. A spherical ball of radius 1×10^{-4} m and density 10^4 kg/m^3 falls freely under gravity through a distance h before entering a tank of water. If after entering the water the velocity of the ball does not change, find h. The viscosity of water is 9.8×10^{-6} N-s/m^2. (g = 9.8 m/s^2)

FM0022

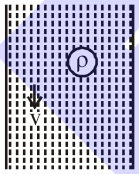
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

56. A spherical ball of density ρ and radius 0.003m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 $N.m^{-2}$ and its density $\rho_L = \rho/2 = 1260\ kg.m^{-3}$. Assume the ball reaches a terminal speed by the 10 cm mark. Find the time taken by the ball to traverse the distance between the 10 cm and 20 cm mark.
(g = acceleration due to gravity = 10 ms^{-2})



FM0023

57. A liquid of density ρ = 1000 kg/m^3 and coefficient of viscosity η = 0.1 Ns/m^2 is flowing down in vertical pipe of large cross section. A small ball of density $\rho_0 = 100\ kg/m^3$ and r = 5 cm will be at rest in flowing liquid, if velocity of flowing liquid is 10 k m/s. Then find the value of k.



FM0024

58. A small sphere falls from rest in a viscous liquid. Due to friction, heat is produced. Find the relation between the rate of production of heat and the radius of the sphere at terminal velocity.

[IIT-JEE 2004]

FM0025

59. When a sphere of radius $r_1 = 1.2$ mm moves in glycerin, the laminar flow is observed if the velocity of the sphere does not exceed $v_1 = 23$ cm/s. At what minimum velocity v_2 of a sphere of radius $r_2 = 5.5$ cm will the flow in water become turbulent? The viscosities of glycerin and water are equal $\eta_1 = 13.9$ P and $\eta_2 = 0.011$ P respectively. ($\rho_g = 1.26$ gm/cc)

FM0373

60. The cross-sectional radius of a pipeline decreases gradually as $r = r_0 e^{-\alpha x}$, where $\alpha = 0.50\ m^{-1}$, x is the distance from the pipeline inlet. Find the ratio of Reynolds numbers $\left(\frac{Re_{final}}{Re_{initial}}\right)$ for two cross-sections separated by Δx = 3.2 m.

FM0374

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

EXERCISE (JM)

1. A uniform cylinder of length L and mass M having cross- sectional area A is suspended, with its length vertical, form a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is : **[AIEEE-2013]**

(1) $\frac{Mg}{k}$ (2) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$ (3) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$ (4) $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

(Here k is spring constant)

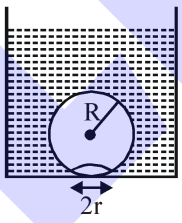
FM0144

2. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporization. **[AIEEE-2013]**

(1) $\frac{\rho L}{T}$ (2) $\sqrt{\frac{T}{\rho L}}$ (3) $\frac{T}{\rho L}$ (4) $\frac{2T}{\rho L}$

FM0145

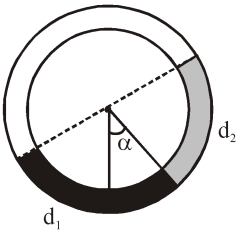
3. On heating water, bubbles being formed at the bottom of the vessel detatch and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If r << R, and the surface tension of water is T, value of r just before bubbles detatch is:- (density of water is ρ_w) **[JEE Mains-2014]**



(1) $R^2\sqrt{\frac{\rho_w g}{T}}$ (2) $R^2\sqrt{\frac{3\rho_w g}{T}}$ (3) $R^2\sqrt{\frac{\rho_w g}{3T}}$ (4) $R^2\sqrt{\frac{\rho_w g}{6T}}$

FM0146

4. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtends 90° angle at centre. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is : **[JEE Mains-2014]**



(1) $\frac{1+\tan\alpha}{1-\tan\alpha}$ (2) $\frac{1+\sin\alpha}{1-\cos\alpha}$ (3) $\frac{1+\sin\alpha}{1-\sin\alpha}$ (4) $\frac{1+\cos\alpha}{1-\cos\alpha}$

FM0147

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

5. Water flows into a large tank with flat bottom at the rate of $10^{-4}\ m^3s^{-1}$. Water is also leaking out of a hole of area 1 cm^2 at its bottom. If the height of the water in the tank remains steady, then this height is: **[JEE-Main-2019_Jan]**

(1) 4 cm (2) 2.9 cm (3) 1.7 cm (4) 5.1 cm **FM0375**

6. A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be: **[JEE-Main-2019_Jan]**

(1) 1.2 (2) 0.1 (3) 2.0 (4) 0.4 **FM0376**

7. If 'M' is the mass of water that rises in a capillary tube of radius 'r', then mass of water which will rise in a capillary tube of radius '2r' is : **[JEE-Main-2019_April]**

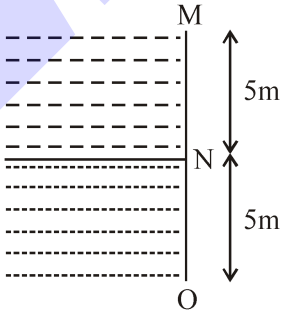
(1) 4M (2) M (3) 2M (4) $\frac{M}{2}$ **FM0377**

8. Consider a solid sphere of radius R and mass density $\rho(r)=\rho_0\left(1-\frac{r^2}{R^2}\right)$, $0<r\le R$. The minimum density of a liquid in which it will float is : **[JEE-Main-2020_Jan]**

(1) $\frac{\rho_0}{5}$ (2) $\frac{\rho_0}{3}$ (3) $\frac{2\rho_0}{3}$ (4) $\frac{2\rho_0}{5}$

FM0378

9. Two liquids of densities ρ_1 an ρ_2 ($\rho_2=2\rho_1$) are filled up behind a square wall of side 10 m as shown in figure. Each liquid has a height of 5 m. The ratio of the forces due to these liquids exerted on upper part MN to that at the lower part NO is (Assume that the liquids are not mixing) **[JEE-Main-2020_Jan]**



(1) 1/4 (2) 2/3 (3) 1/3 (4) 1/2 **FM0379**

10. A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet) : **[JEE-Main-2020_Jan]**

(1) $r=\sqrt{\frac{2T}{3(d+\rho)g}}$ (2) $r=\sqrt{\frac{3T}{(2d-\rho)g}}$ (3) $r=\sqrt{\frac{T}{(d-\rho)g}}$ (4) $r=\sqrt{\frac{T}{(d+\rho)g}}$

FM0380

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

11. A leak proof cylinder of length 1m, made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm. When the temperature of water is increased to 4°C, the height of the cylinder above the water surface becomes 21 cm. The density of water at T = 4°C, relative to the density at T = 0°C is close to : **[JEE-Main-2020_Jan]**

(1) 1.01 (2) 1.04 (3) 1.03 (4) 1.26

FM0381

12. A air bubble of radius 1 cm in water has an upward acceleration 9.8 cm s^{-2}. The density of water is 1 gm cm^{-3} and water offers negligible drag force on the bubble. The mass of the bubble is ($g = 980\ cm/s^2$) **[JEE-Main-2020_Sep]**

(1) 3.15 gm (2) 4.51 gm (3) 4.15 gm (4) 1.52 gm

FM0382

13. A fluid is flowing through a horizontal pipe of varying cross-section, with speed v ms^{-1} at a point where the pressure is P Pascal. P At another point where pressure is $\frac{P}{2}$ Pascal its speed is V ms^{-1}. If the density of the fluid is ρ kg m^{-3} and the flow is streamline, then V is equal to : **[JEE-Main-2020_Sep]**

(1) $\sqrt{\frac{P}{2\rho}+v^2}$ (2) $\sqrt{\frac{P}{\rho}+v^2}$ (3) $\sqrt{\frac{2P}{\rho}+v^2}$ (4) $\sqrt{\frac{P}{\rho}+v}$

FM0383

14. A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be: **[JEE-Main-2021_Feb]**

(1) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$ (2) $\frac{2T}{rJ}$ (3) $\frac{3T}{rJ}$ (4) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

FM0384

15. What will be the nature of flow of water from a circular tap, when its flow rate increased from 0.18 L/min to 0.48 L/min ? The radius of the tap and viscosity of water are 0.5 cm and 10^{-3} Pa s, respectively. (Density of water : 10^3 kg/m^3) **[JEE-Main-2021_March]**

(1) Unsteady to steady flow (2) Remains steady flow

(3) Remains turbulent flow (4) Steady flow to unsteady flow **FM0385**

16. Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}}\times10^{-3}$ cm. Then the thickness of oleic acid layer will be $x \times 10^{-14}$ m. Where x is_____.

[JEE-Main-2021_March]

FM0386

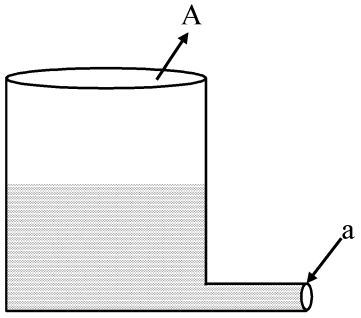
17. Two spherical soap bubbles of radii r_1 and r_2 in vacuum combine under isothermal conditions. The resulting bubble has a radius equal to : **[JEE-Main-2021_July]**

(1) $\frac{r_1r_2}{r_1+r_2}$ (2) $\sqrt{r_1r_2}$ (3) $\sqrt{r_1^2+r_2^2}$ (4) $\frac{r_1+r_2}{2}$

FM0387

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

18. A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is (a << A) : **[JEE-Main-2021_July]**



(1) $\frac{A}{2a}$ (2) None of these (3) $\frac{2a}{A}$ (4) $\frac{a}{A}$

FM0388

19. Two narrow bores of diameter 5.0 mm and 8.0 mm are joined together to form a U-shaped tube open at both ends. If this U-tube contains water, what is the difference in the level of two limbs of the tube. [Take surface tension of water T = 7.3 × 10^{-2} Nm^{-1}, angle of contact = 0, g = 10 ms^{-2} and density of water = 1.0 × 10^3 kg m^{-3}] **[JEE-Main-2021_Aug]**

(1) 3.62 mm (2) 2.19 mm (3) 5.34 mm (4) 4.97 mm

FM0389

20. A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C. What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is –7°C. Assuming the volume constant ? **[JEE-Main-2021_Aug]**

(1) 181.46 kg (2) 214.15 kg. (3) 219.07 kg (4) 123.54 kg

FM0390

21. The velocity of a small ball of mass 'm' and density d_1, when dropped in a container filled with glycerine, becomes constant after some time. If the density of glycerine is d_2, then the viscous force acting on the ball, will be : **[JEE-Main-2022_June]**

(1) $mg\left(1-\frac{d_1}{d_2}\right)$ (2) $mg\left(1-\frac{d_2}{d_1}\right)$ (3) $mg\left(\frac{d_1}{d_2}-1\right)$ (4) $mg\left(\frac{d_2}{d_1}-1\right)$

FM0391

22. A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension 7.5 × 10^{-4} Ncm^{-1}. The radius of drop in cm will be : (Take : g = 10 m/s^2) **[JEE-Main-2022_July]**

(1) $\frac{15}{\sqrt{2\rho-\sigma}}$ (2) $\frac{15}{\sqrt{\rho-\sigma}}$ (3) $\frac{3}{2\sqrt{\rho-\sigma}}$ (4) $\frac{3}{20\sqrt{2\rho-\sigma}}$

FM0392

23. A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from $A_1 = 1.2\times10^{-2}\,m^2$ to $A_2 = \frac{A_1}{2}$. The pressure difference between the wide and narrow sections of the pipe is 4500 Pa. The rate of flow of liquid is _____ $\times10^{-3}\,m^3s^{-1}$. **[JEE-Main-2022_June]**

FM0393

EXERCISE (JA)

1. A solid sphere of radius R and density ρ is attached to one end of a mass-less spring of force constant k. The other end of the spring is connected to another solid sphere of radius R and density 3ρ. The complete arrangement is placed in a liquid of density 2ρ and is allowed to reach equilibrium. The correct statement(s) is (are) **[JEE-Advance-2013]**

(A) the net elongation of the spring is $\frac{4\pi R^3\rho g}{3k}$.

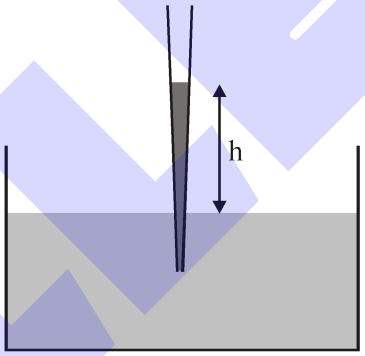
(B) the net elongation of the spring is $\frac{8\pi R^3\rho g}{3k}$.

(C) the light sphere is partially submerged.

(D) the light sphere is completely submerged. **FM0156**

2. A glass capillary tube is of the shape of truncated cone with an apex angle α so that its two ends have cross sections of different radii. When dipped in water vertically, water rises in it to height h, where the radius of its cross section is b. If the surface tension of water is S, its density is ρ, and its contact angle with glass is θ, the value of h will be (g is the acceleration due to gravity)

[JEE Advanced-2014]

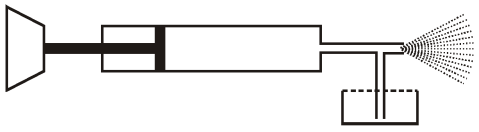


(A) $\frac{2S}{b\rho g}\cos(\theta-\alpha)$ (B) $\frac{2S}{b\rho g}\cos(\theta+\alpha)$ (C) $\frac{2S}{b\rho g}\cos(\theta-\alpha/2)$ (D) $\frac{2S}{b\rho g}\cos(\theta+\alpha/2)$

FM0157

Paragraph for Questions 3 & 4

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere. **[JEE Advanced-2014]**



3. If the piston is pushed at a speed of 5 mms^{-1}, the air comes out of the nozzle with a speed of

(A) 0.1 ms^{-1} (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 8 ms^{-1}

FM0158

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

4. If the density of air is ρ_a and that of the liquid ρ_ℓ, then for a given piston speed the rate (volume per unit time) at which the liquid is sprayed will be proportional to

(A) $\sqrt{\frac{\rho_a}{\rho_\ell}}$ (B) $\sqrt{\rho_a\rho_\ell}$ (C) $\sqrt{\frac{\rho_\ell}{\rho_a}}$ (D) $\rho\ell$

FM0158

5. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following, state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists. **[JEE Advanced-2014]**

List - I		**List-II**	
(P)	Lift is accelerating vertically up.	(1)	d = 1.2 m
(Q)	Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.	(2)	d > 1.2 m
		(3)	d < 1.2 m
(R)	Lift is moving vertically up with constant speed.	(4)	No water leaks out of the jar
(S)	Lift is falling freely.		

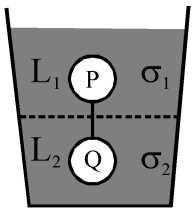
Code :

(A) P-2, Q-3, R-2, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-1, Q-1, R-1, S-4 (D) P-2, Q-3, R-1, S-1 **FM0159**

6. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2, respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2, respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{V}_P$ and Q alone in L_1 has terminal velocity $\vec{V}_Q$, then

[JEE Advanced-2015]



(A) $\frac{|\vec{V}_P|}{|\vec{V}_Q|}=\frac{\eta_1}{\eta_2}$ (B) $\frac{|\vec{V}_P|}{|\vec{V}_Q|}=\frac{\eta_2}{\eta_1}$ (C) $\vec{V}_P.\vec{V}_Q>0$ (D) $\vec{V}_P.\vec{V}_Q<0$

FM0160

7. Consider two solid spheres P and Q each of density 8 gm cm^{-3} and diameters 1 cm and 0.5cm, respectively. Sphere P is dropped into a liquid of density 0.8 gm cm^{-3} and viscosity η = 3 poiseulles. Sphere Q is dropped into a liquid of density 1.6 gm cm^{-3} and viscosity η = 2 poiseulles. The ratio of the terminal velocities of P and Q is. **[JEE Advanced-2016]**

FM0161

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

8. A drop of liquid of radius R = 10^{-2} m having surface tension S = $\frac{0.1}{4\pi}$ Nm^{-1} divides itself into K identical drops. In this process the total change in the surface energy $\Delta U = 10^{-3}$ J. If K = 10^{α} then the value of α is **[JEE Advanced-2017]**

FM0162

9. A uniform capillary tube of inner radius r is dipped vertically into a beaker filled with water. The water rises to a height h in the capillary tube above the water surface in the beaker. The surface tension of water is σ. The angle of contact between water and the wall of the capillary tube is θ. Ignore the mass of water in the meniscus. Which of the following statements is (are) true? **[JEE Advanced-2018]**

(A) For a given material of the capillary tube, h decreases with increase in r
(B) For a given material of the capillary tube, h is independent of σ.
(C) If this experiment is performed in a lift going up with a constant acceleration, then h decreases.,
(D) h is proportional to contact angle θ. **FM0163**

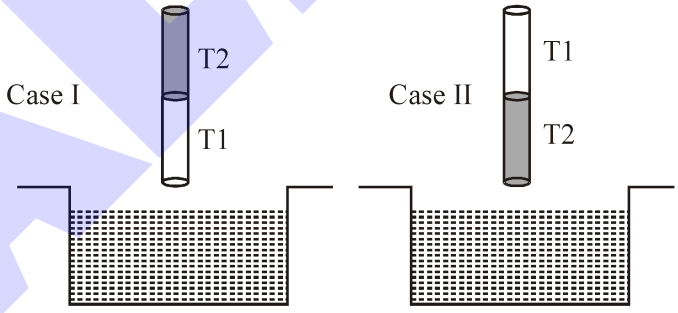
10. Consider a thin square plate floating on a viscous liquid in a large tank. The height h of the liquid in the tank is much less than the width of the tank. The floating plate is pulled horizontally with a constant velocity u_0. Which of the following statements is (are) true ? **[JEE Advanced-2018]**

(A) The resistive force of liquid on the plate is inversely proportional to h
(B) The resistive force of liquid on the plate is independent of the area of the plate
(C) The tangential (shear) stress on the floor of the tank increases with u_0.
(D) The tangential (shear) stress on the plate varies linearly with the viscosity η of the liquid.

FM0164

11. A cylindrical capillary tube of 0.2 mm radius is made by joining two capillaries T1 and T2 of different materials having water contact angles of 0° and 60°, respectively. The capillary tube is dipped vertically in water in two different configurations, case I and II as shown in figure. Which of the following option(s) is(are) correct ?
(Surface tension of water = 0.075 N/m, density of water = 1000 kg/m^3, take g = 10 m/s^2)

[JEE Advanced-2019]

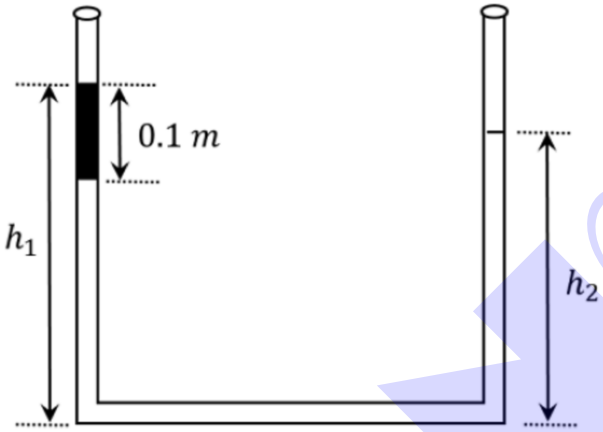


(A) The correction in the height of water column raised in the tube, due to weight of water contained in the meniscus, will be different for both cases.
(B) For case I, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be more than 8.75 cm. (Neglect the weight of the water in the meniscus)
(C) For case I, if the joint is kept at 8 cm above the water surface, the height of water column in the tube will be 7.5 cm. (Neglect the weight of the water in the meniscus)
(D) For case II, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be 3.75 cm. (Neglect the weight of the water in the meniscus)

FM0165

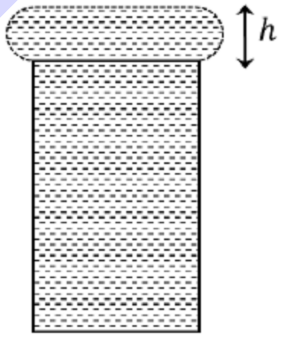
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

12. An open-ended U-tube of uniform cross-sectional area contains water (density 10^3kg m^{-3}). Initially the water level stands at 0.29 m from the bottom in each arm. Kerosene oil (a water-immiscible liquid) of density 800 kg m^{-3} is added to the left arm until its length is 0.1 m, as shown in the schematic figure below. The ratio $\left(\frac{h_1}{h_2}\right)$ of the heights of the liquid in the two arms is- **[JEE Advanced-2020]**



(A) $\frac{15}{14}$ (B) $\frac{35}{33}$ (C) $\frac{7}{6}$ (D) $\frac{5}{4}$ **FM0394**

13. When water is filled carefully in a glass, one can fill it to a height h above the rim of the glass due to the surface tension of water. To calculate h just before water starts flowing, model the shape of the water above the rim as a disc of thickness h having semicircular edges, as shown schematically in the figure. When the pressure of water at the bottom of this disc exceeds what can be withstood due to the surface tension, the water surface breaks near the rim and water starts flowing from there. If the density of water, its surface tension and the acceleration due to gravity are 10^3 kg m^{-3}, 0.07 Nm^{-1} and 10 ms^{-2}, respectively, the value of h (in mm) is _____.- **[JEE Advanced-2020]**



FM0395

14. A train with cross-sectional area S_t is moving with speed v_t inside a long tunnel of cross-sectional area S_0 ($S_0 = 4S_t$). Assume that almost all the air (density ρ) in front of the train flows back between its sides and the walls of the tunnel. Also, the air flow with respect to the train is steady and laminar. Take the ambient pressure and that inside the train to be p_0. If the pressure in the region between the sides of the train and the tunnel walls is p, then $p_0 - p = \frac{7}{2N}\rho v_t^2$. The value of N is _____. **[JEE Advanced-2020]**

FM0396

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4 Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

15. A hot air balloon is carrying some passengers, and a few sandbags of mass 1 kg each so that its total mass is 480 kg. Its effective volume giving the balloon its buoyancy is V. The balloon is floating at an equilibrium height of 100 m. When N number of sandbags are thrown out, the balloon rises to a new equilibrium height close to 150 m with its volume V remaining unchanged. If the variation of the density of air with height h from the ground is $\rho(h) = \rho_0 e^{-\frac{h}{h_0}}$, where $\rho_0 = 1.25$ kg m^{-3} and $h_0 = 6000$ m, the value of N is _____.

[JEE Advanced-2020]

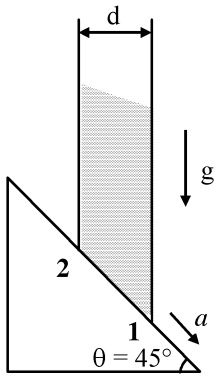
FM0397

16. A cubical solid aluminium (bulk modulus $= -V\frac{dP}{dV} = 70$ GPa) block has an edge length of 1 m on the surface of the earth. It is kept on the floor of a 5 km deep ocean. Taking the average density of water and the acceleration due to gravity to be 10^3 kg m^{-3} and 10 ms^{-2}, respectively, the change in the edge length of the block in mm is _____.

[JEE Advanced-2020]

FM0398

17. A cylindrical tube, with its base as shown in the figure, is filled with water. It is moving down with a constant acceleration a along a fixed inclined plane with angle $\theta = 45°$. P_1 and P_2 are pressures at points **1** and **2**, respectively, located at the base of the tube. Let $\beta = (P_1 - P_2)/(\rho g d)$, where ρ is density of water, d is the inner diameter of the tube and g is the acceleration due to gravity. Which of the following statement(s) is(are) correct ?



[JEE Advanced-2021]

(A) $\beta = 0$ when $a = g/\sqrt{2}$

(B) $\beta > 0$ when $a = g/\sqrt{2}$

(C) $\beta = \frac{\sqrt{2}-1}{\sqrt{2}}$ when $a = g/2$

(D) $\beta = \frac{1}{\sqrt{2}}$ when $a = g/2$

FM0399

Paragraph for Questions 18 and 19

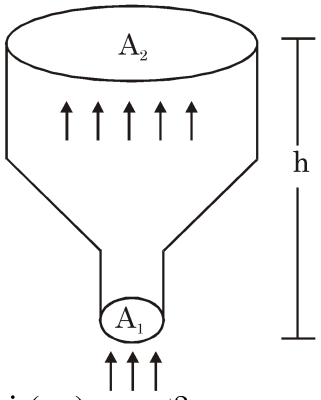
A soft plastic bottle, filled with water of density 1 gm/cc, carries an inverted glass test-tube with some air (ideal gas) trapped as shown in the figure. The test-tube has a mass of 5 gm, and it is made of a thick glass of density 2.5 gm/cc. Initially the bottle is sealed at atmospheric pressure $p_0 = 10^5$ Pa so that the volume of the trapped air is $v_0 = 3.3$ cc. When the bottle is squeezed from outside at constant temperature, the pressure inside rises and the volume of the trapped air reduces. It is found that the test tube begins to sink at pressure $P_0 + \Delta p$ without changing its orientation. At this pressure, the volume of the trapped air is $v_0 - \Delta v$. Let $\Delta v = X$ cc and $\Delta p = Y \times 10^3$ Pa. **[JEE-Advance 2021]**

18. The value of X is ____. **FM0400**

19. The value of Y is ____. **FM0400**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

20. An ideal gas of density $\rho = 0.2$ kg m^{-3} enters a chimney of height h at the rate of $\alpha = 0.8$ kg s^{-1} from its lower end, and escapes through the upper end as shown in the figure. The cross-sectional area of the lower end is $A_1 = 0.1$ m^2 and the upper end is $A_2 = 0.4$ m^2. The pressure and the temperature of the gas at the lower end are 600 Pa and 300 K, respectively, while its temperature at the upper end is 150 K. The chimney is heat insulated so that the gas undergoes adiabatic expansion. Take $g = 10$ ms^{-2} and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure. **[JEE Advanced-2022]**



Which of the following statement(s) is(are) correct?

(A) The pressure of the gas at the upper end of the chimney is 300 Pa.

(B) The velocity of the gas at the lower end of the chimney is 40 ms^{-1} and at the upper end is 20 ms^{-1}.

(C) The height of the chimney is 590 m.

(D) The density of the gas at the upper end is 0.05 kg m^{-3}. **FM0401**

21. A bubble has surface tension S. The ideal gas inside the bubble has ratio of specific heats $\gamma = \frac{5}{3}$. The bubble is exposed to the atmosphere and it always retains its spherical shape. When the atmospheric pressure is P_{a1}, the radius of the bubble is found to be r_1 and the temperature of the enclosed gas is T_1. When the atmospheric pressure is P_{a2}, the radius of the bubble and the temperature of the enclosed gas are r_2 and T_2, respectively. Which of the following statement(s) is(are) correct?

[JEE Advanced-2022]

(A) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{r_1}{r_2}\right)^5 = \frac{P_{a2} + \frac{2S}{r_2}}{P_{a1} + \frac{2S}{r_1}}$

(B) If the surface of the bubble is a perfect heat insulator, then the total internal energy of the bubble including its surface energy does not change with the external atmospheric pressure.

(C) If the surface of the bubble is a perfect heat conductor and the change in atmospheric temperature is negligible, then $\left(\frac{r_1}{r_2}\right)^3 = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$.

(D) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{T_2}{T_1}\right)^{\frac{5}{2}} = \frac{P_{a2} + \frac{4S}{r_2}}{P_{a1} + \frac{4S}{r_1}}$. **FM0402**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.pt5

ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	A	A	A	D	A	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	C	B	A	C	C	D	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	D	B	A	B	C	C	A	C
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	A	C	C	B	B	C	B	D	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	B	A	B	B	C	D	A	D	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	D	A	D	D	A	B	C	C	A
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	B	D	C	B	B	B	B	B	C	C
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	A	C	B	B	C	B	C	C	C	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	B	D	A	D	B	B	C	A	C
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	B	C	B	C	B	A	D	C	A	D
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	A	C	B	B	D	A	A	B	B	A
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	A	A	C	D	A	B	B	D	A	B
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	B	A	B	B	C	D	B	B	A	C
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	D	B	A	D	A	C	B	B	C	A
Que.	141	142	143	144	145	146	147	148	149	150
Ans.	B	C	A	C	D	B	B	A	C	B
Que.	151	152	153	154	155	156	157	158	159	160
Ans.	D	B	C	C	B	C	B	C	A	B
Que.	161	162	163	164	165	166	167	168	169	170
Ans.	B	A	A,C	A,D	B,D	A,B,C	B,C	A,C	A,B	C,D
Que.	171	172	173	174	175	176	177	178	179	
Ans.	C,D	B,C,D	C,D	A,B,C	D	B	B	A-R; B-Q; C-Q; D-Q	A-P,T; B-P,S; C-Q,T; D-R	

node06\B0BC-BD\Kota \JEE(Advanced)\Nurture \Phy \Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	C	A	C	A,B,C	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	A	A	B	B	C	C	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	D	D	D	D	C	D	A	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	D	D	A,C	B,C,D	C	A	B	C	B
Que.	41	42	43							
Ans.	A	A-Q; B-R; C-P; D-S	A-P,T; B-Q,R,S; C-Q,R,S,T; D-P							

EXERCISE # S

1. $H = \frac{L^2\omega^2}{2g}$ **2. (10 sec.)** **3. nH** **4. 0.1 cm** **5. 3 : 9 : 5** **6. 8**

7. (a) 60000 N, **(b)** 20000 x δ x N, **(c)** 20000 x (1 – x)δx N-m, **(d)** 10000 N, **(e)** 10000/3 N-m

8. (a) $\left[\frac{dgRH^2}{3}\right]$, (b) $\frac{dg\pi R^2H}{6}$ **9.** 45°, 9600 $\sqrt{2}$ Pa **10.** $P_0 - \rho a\ell$ **11.** $\sqrt{\frac{18g}{19a}}$

12. (a) The paraboloid of revolution ; $z = \left(\frac{\omega^2r^2}{2g}\right)$, where z is the height measured from the surface of the liquid along the axis of vessel, r is the distance from the rotation axis ; (b) $p = p_0 + \frac{1}{2}\rho\omega^2r^2$.

13. (a) 204 N; **(b)** 1 N upward **14.** 37.5 N **15. (a) 5, (b) 2/3** **16. 1.9**

17. 10.4 cm **18. (a)** $\frac{1}{20}$ **(b)** $\frac{19}{20}$ **19.** 4.5m **20.** 45° **21. 110**

22. $\frac{Agh^2(\rho_0 - \rho_1)^2}{2\rho_0}$ **23.** 4 **24. 5** **25.** $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$ **26.** h = 5m

27. $\frac{\lambda a}{2y}g$ **28. (a) 4.8 mm; (b)** $\theta = \cos^{-1}\left(\frac{1}{2\sqrt{2}}\right)$ **29.** 45 N **30. (a) 1.5 cm (b) 60°**

31. 5.77 cm **32. 87.2** **33.** $\alpha = \left[\rho gh + \frac{p_0h}{(\ell - h)}\right]\frac{d}{4\cos\theta}$ **34. 1.59**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65

36. 108 **37. 2** **38.** 6.43×10^{-4} m³/s **39. (a)** 25 cm/s; **(b)** 50 cm/s; **(c)** 94 N/m²

40. (a) 25 cm/s; **(b)** 50 cm/s; **(d)** zero **41. (a)** 25 cm/s; **(b)** 50 cm/s; **(c)** 188 N/m² **42. 2.08 cm**

43. 20 N **44.** 11 m/s **45.** + 29625 J/m³, – 30000 J/m³ **46.** x = 2 m

47. 6 **48.** $v = \sqrt{2g\left[h_1 + h_2\left(\frac{\rho_2}{\rho_1}\right)\right]}$ **49. (a)** 10 m/s; **(b)** 2.5 m/s; **(c)** 4.688 m; **(d)** 3.14×10^{-3} m³/s

50. 1890 cm³/s. **51. 6** **52. 4**

53. (a) $Q = \frac{1}{2}\pi v_0 R^2$; **(b)** $T = \frac{1}{6}\rho \ell R^2 \pi v_0^2$; **(c)** $F_{fr} = 4\pi\eta\ell v_0$; **(d)** $\Delta p = 4\eta\ell v_0/R^2$ **54. 8/sec, 4 N/m²**

55. 20.4 m **56. 5 s** **57. 5** **58.** $\frac{dQ}{dt} \propto r^5$ **59. 5 μ m/s**

60. $e^{\alpha\Delta x} = 5$

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	3	1	4	3	3	4	1	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	4	4	25	3	3	2	4
Que.	21	22	23							
Ans.	2	1	24							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,D	D	C	A	C	A,D	3	6	A,C	A,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C,D	B	3.65 to 3.85	9	4	0.23 to 0.25 OR –0.23 to –0.25	A,C	0.30	10.00	B
Que.	21									
Ans.	C,D									

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\4-Fluid Mechanics & SHM\Eng\01_Fluid Mechanics\01_Eng.p65